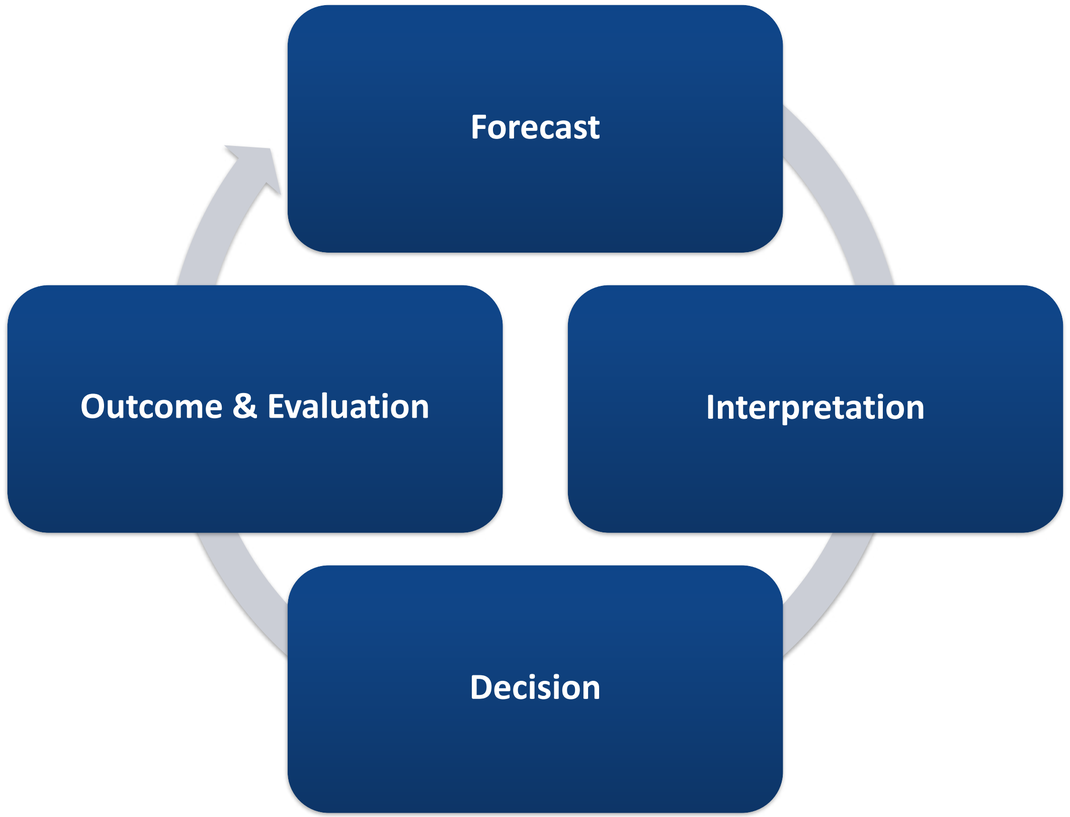
Need for modeling uncertainty in energy forecasting

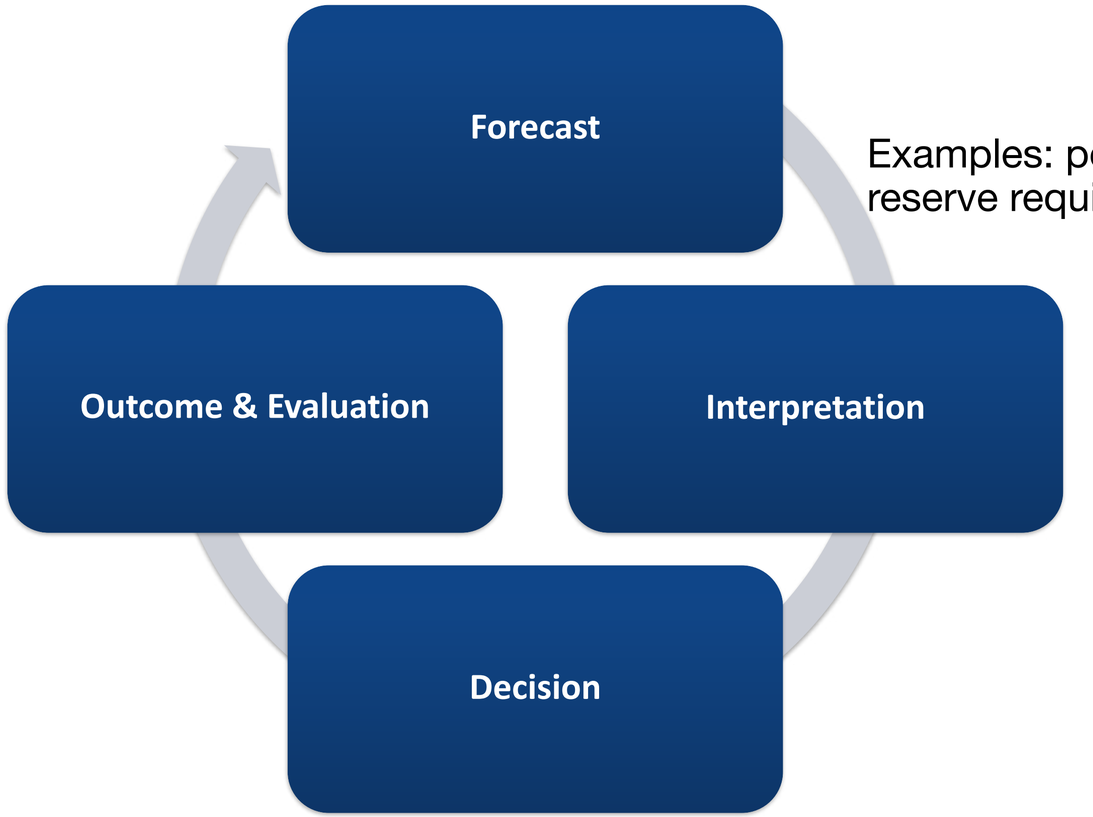
How to make and evaluate forecasts with uncertainty quantification

Hussain Kazmi, PhD
ELECTA-ESAT and EnergyVille
17/03/2026

Forecasting as decision support

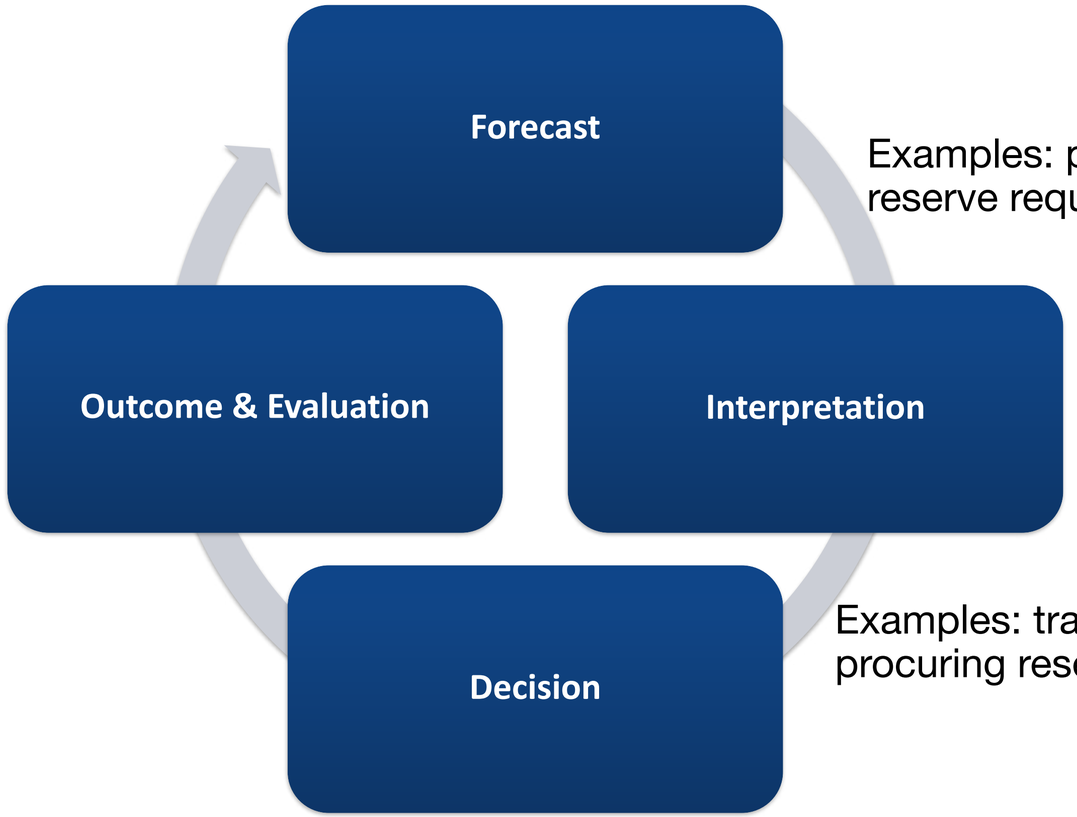


Forecasting as decision support



Examples: power generation / demand forecasts; reserve requirements

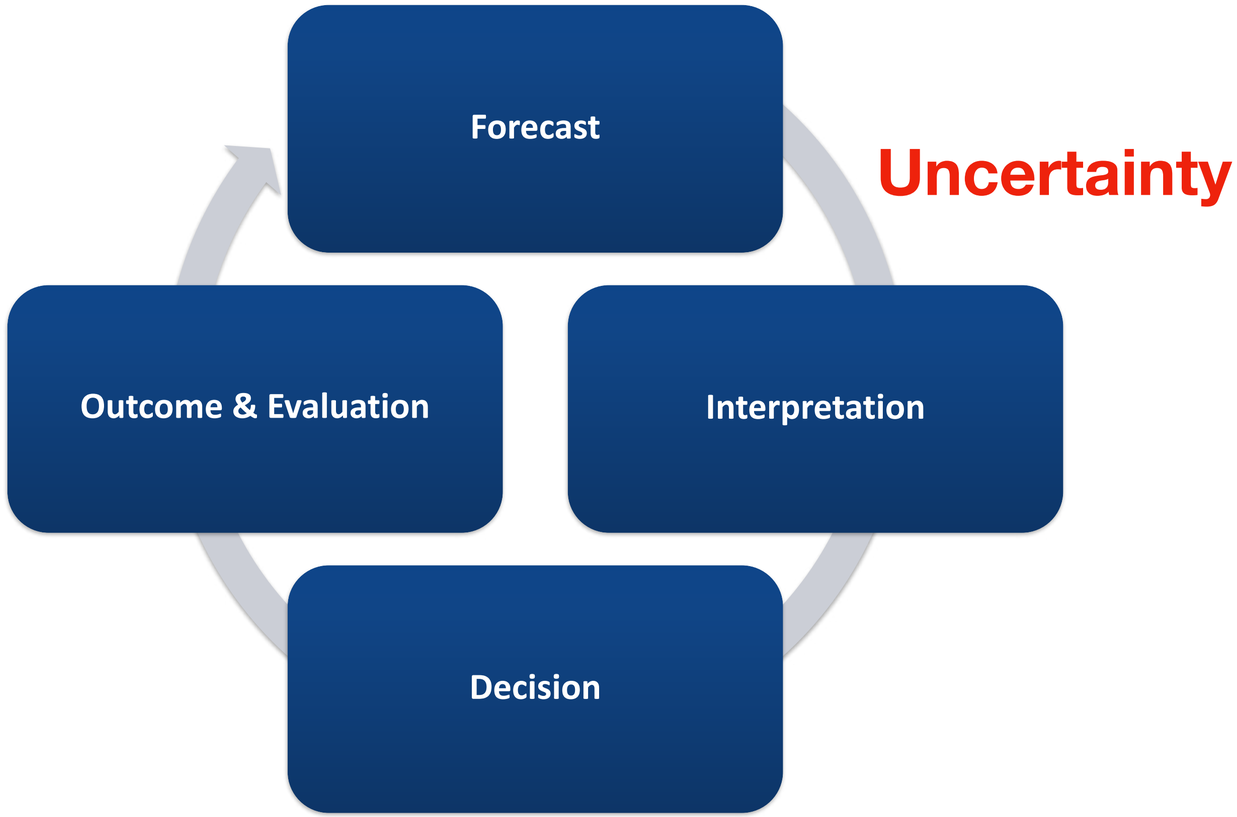
Forecasting as decision support



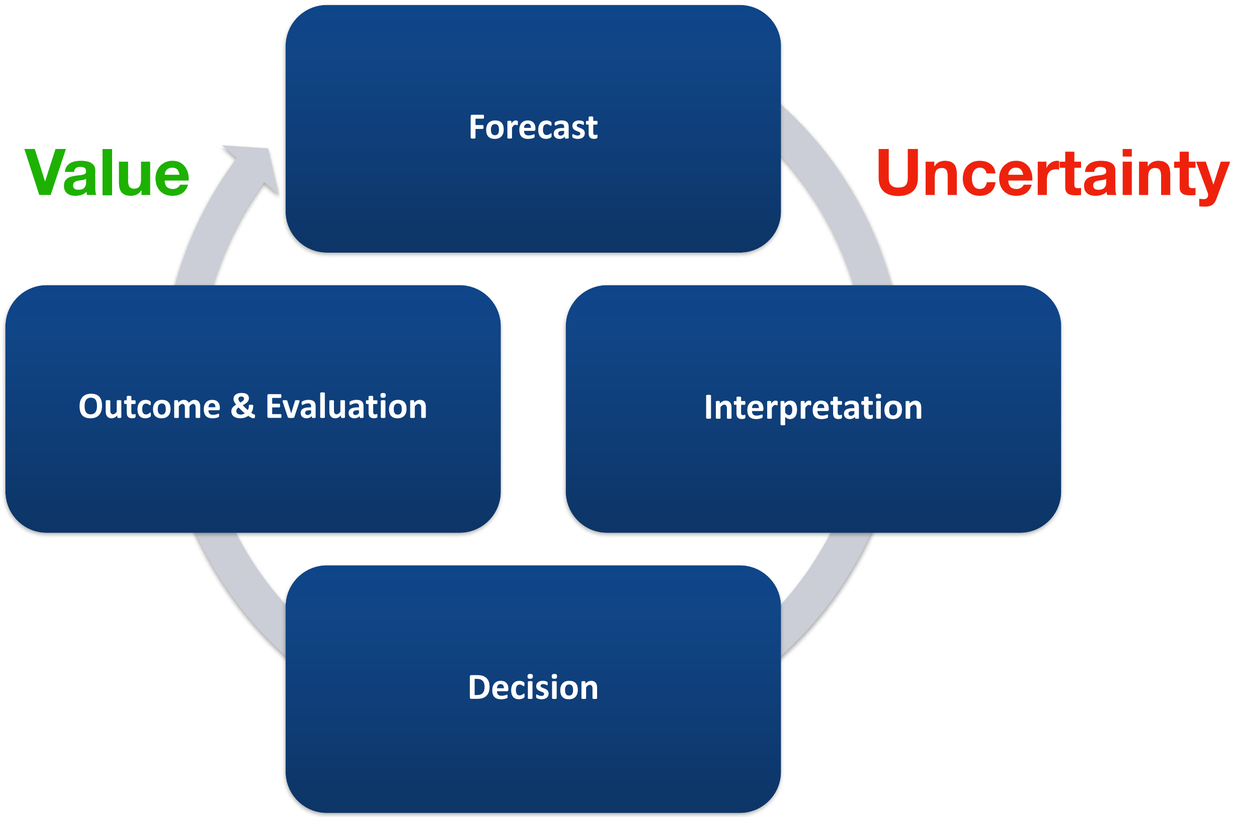
Examples: power generation / demand forecasts; reserve requirements

Examples: trading decisions; procuring reserves

Forecasting as decision support



Forecasting as decision support



What causes uncertainty in the energy sector?

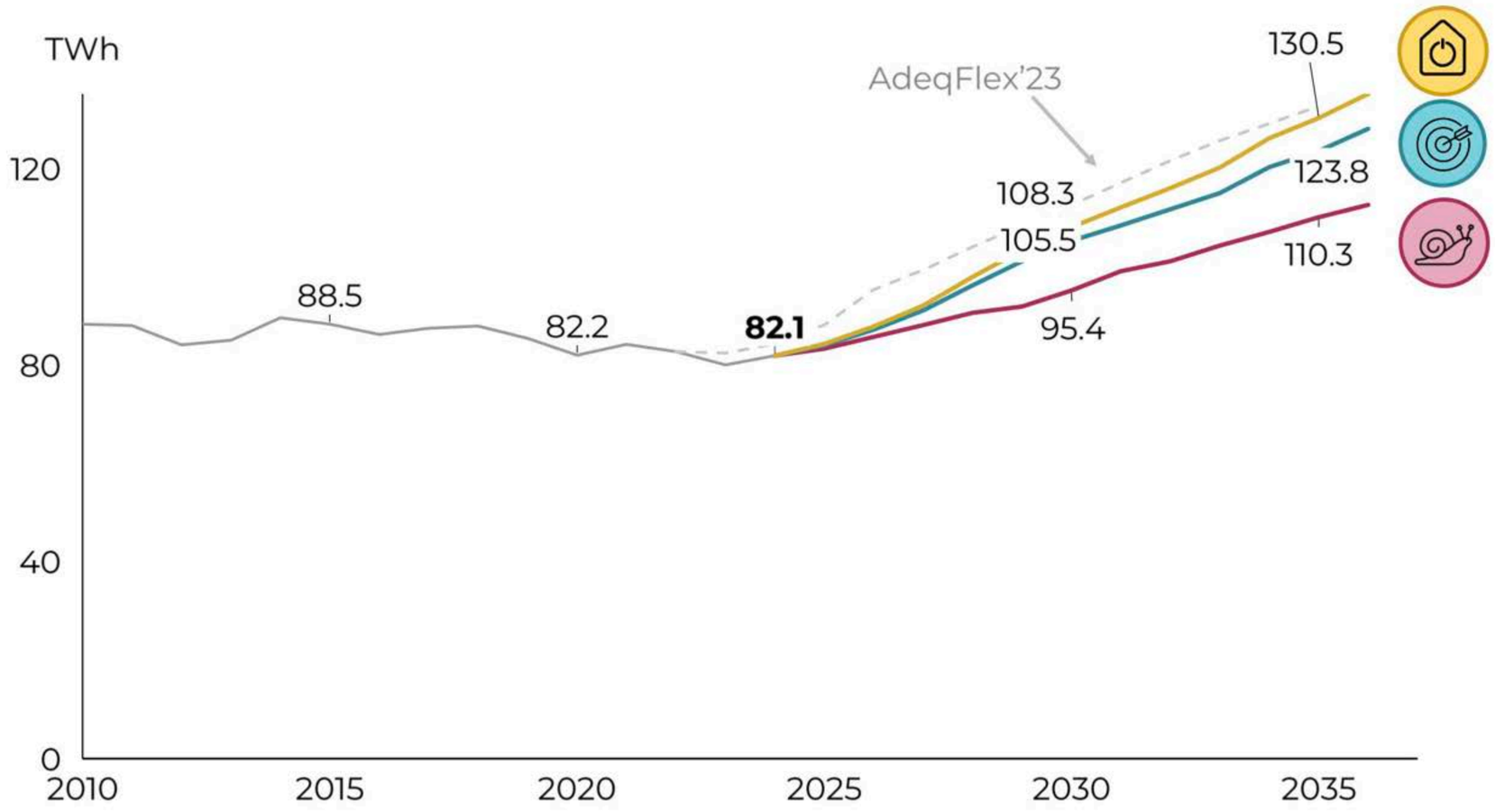
What *else* causes uncertainty?

- In the long term (investment / planning horizon)
  - Socioeconomic factors (e.g. adoption curves, NIMBY-ism …)
  - Technological changes (e.g. new technologies, technological improvement)

What *else* causes uncertainty?

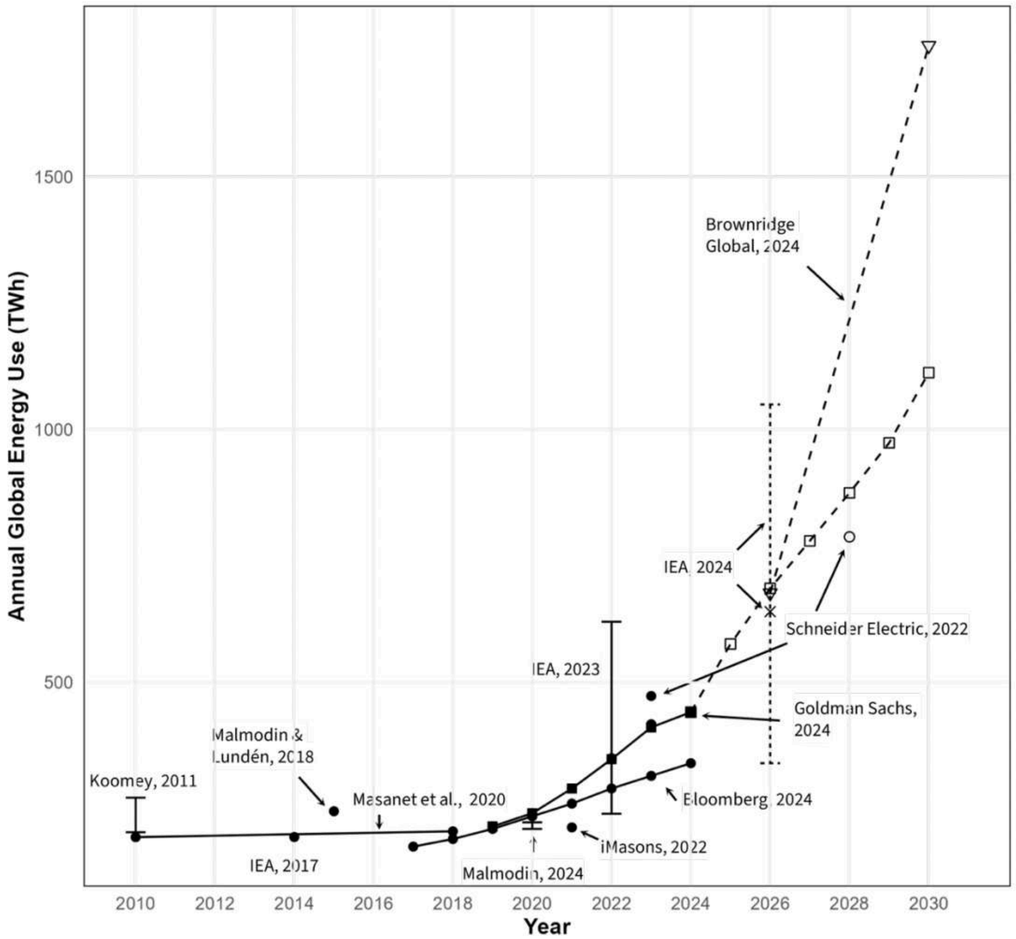
- In the long term (investment / planning horizon)
  - Socioeconomic factors (e.g. adoption curves, NIMBY-ism …)
  - Technological changes (e.g. new technologies, technological improvement
- In the short term (operational horizon)
  - Operational constraints (grid infrastructure limits)
  - Resource (un)availability (e.g. unexpected outages, renewable volatility / intermittency)
  - Lack of information

Some long-term examples (BE load)



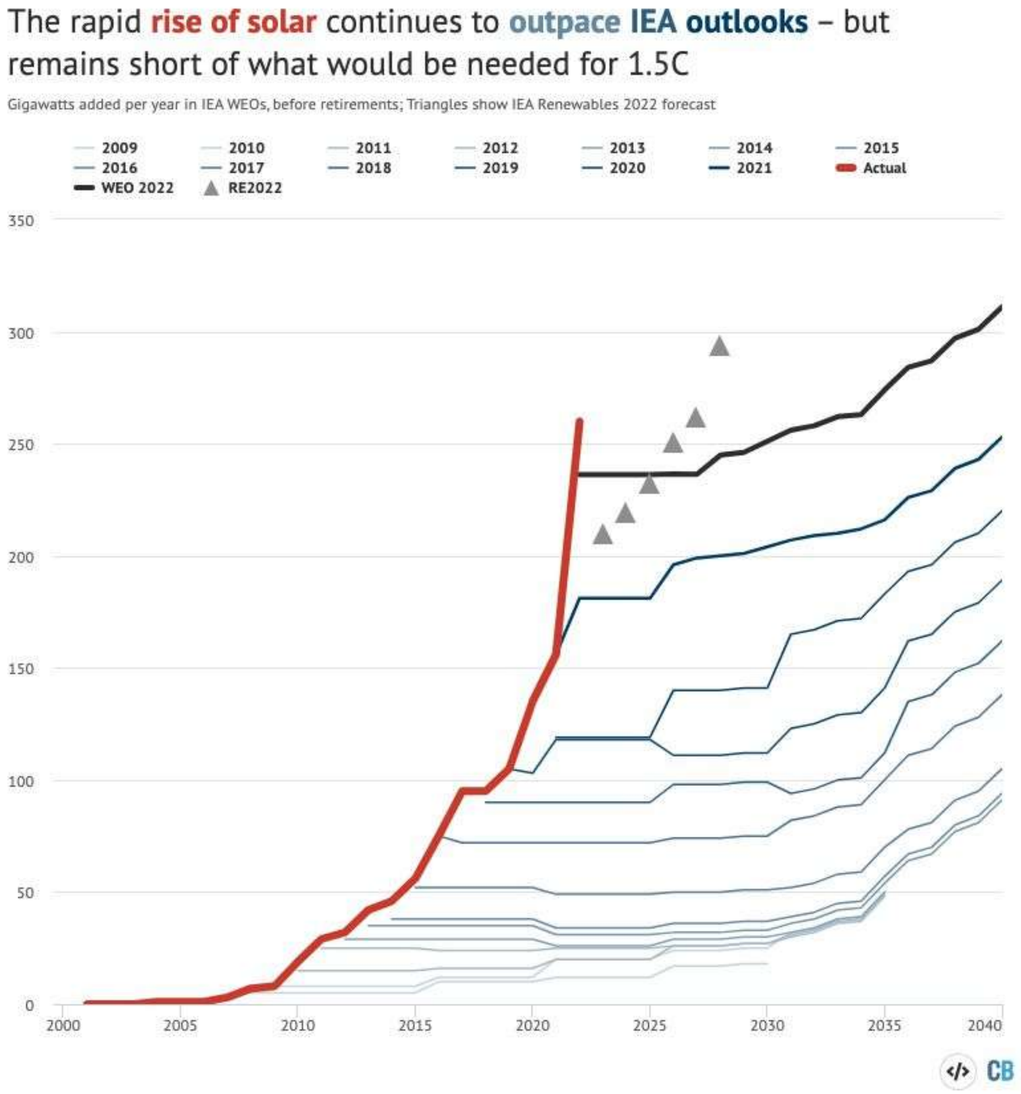
Elia. Adequacy and Flexibility Study. 2026 - 2036

Some long-term examples (data centers)



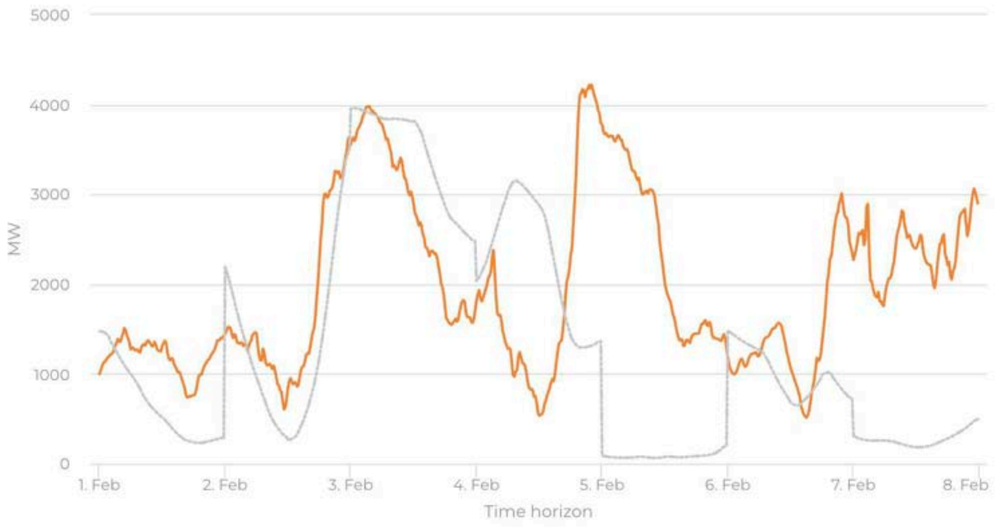
Kim, Jinsoo, et al. "Reviewing the socio-technical dynamics of AI, data centers and digitalization on energy and the environment." *Renewable and Sustainable Energy Reviews* 234 (2026): 116861.

Some long-term examples (IEA PV case)



Simon Evans, 2023.

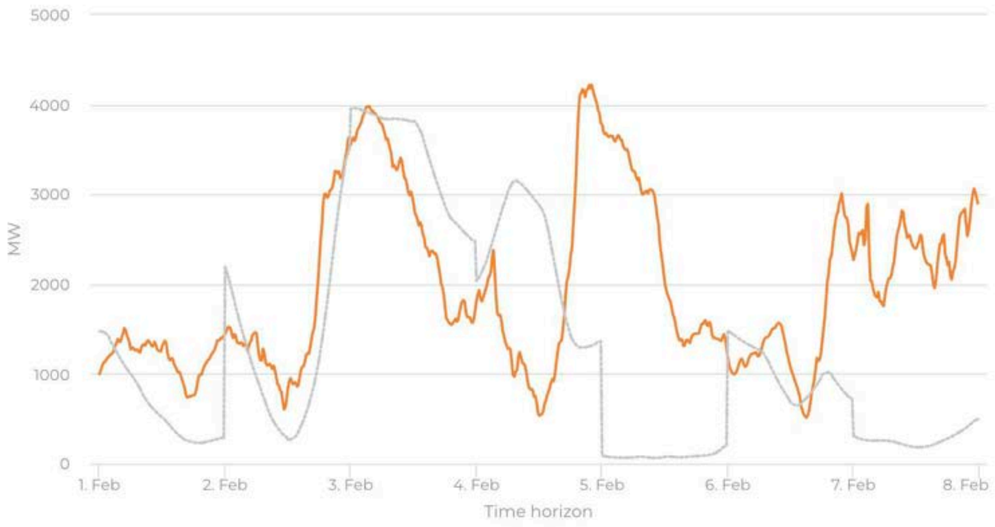
Some short-term examples (wind power - BE)



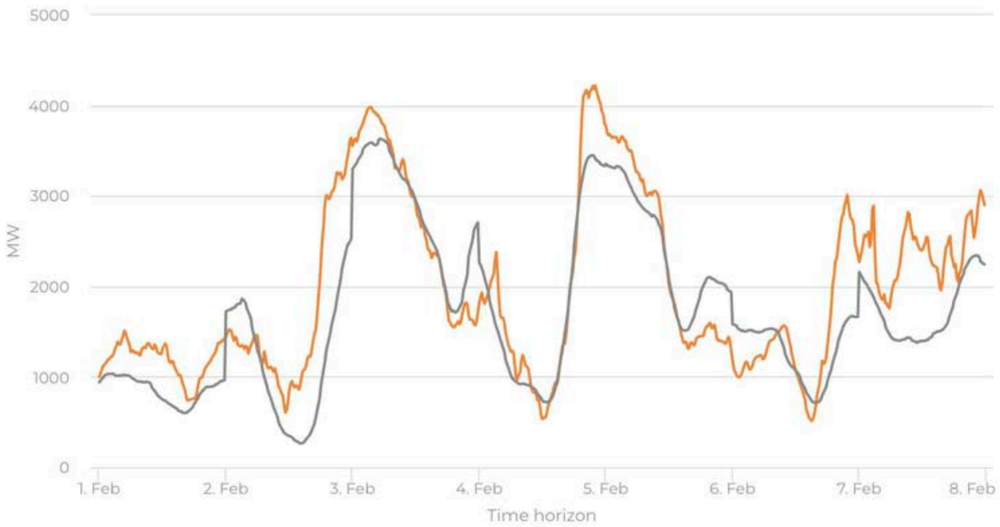
Week-ahead forecast

Elia. Open Data Platform.

Some short-term examples (wind power - BE)



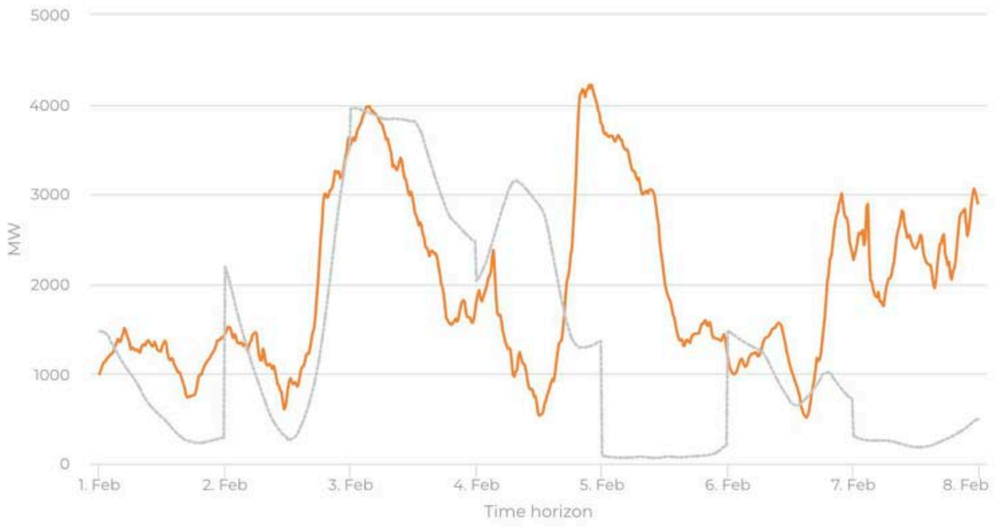
Week-ahead forecast



Day-ahead forecast

Elia. Open Data Platform.

Some short-term examples (wind power - BE)



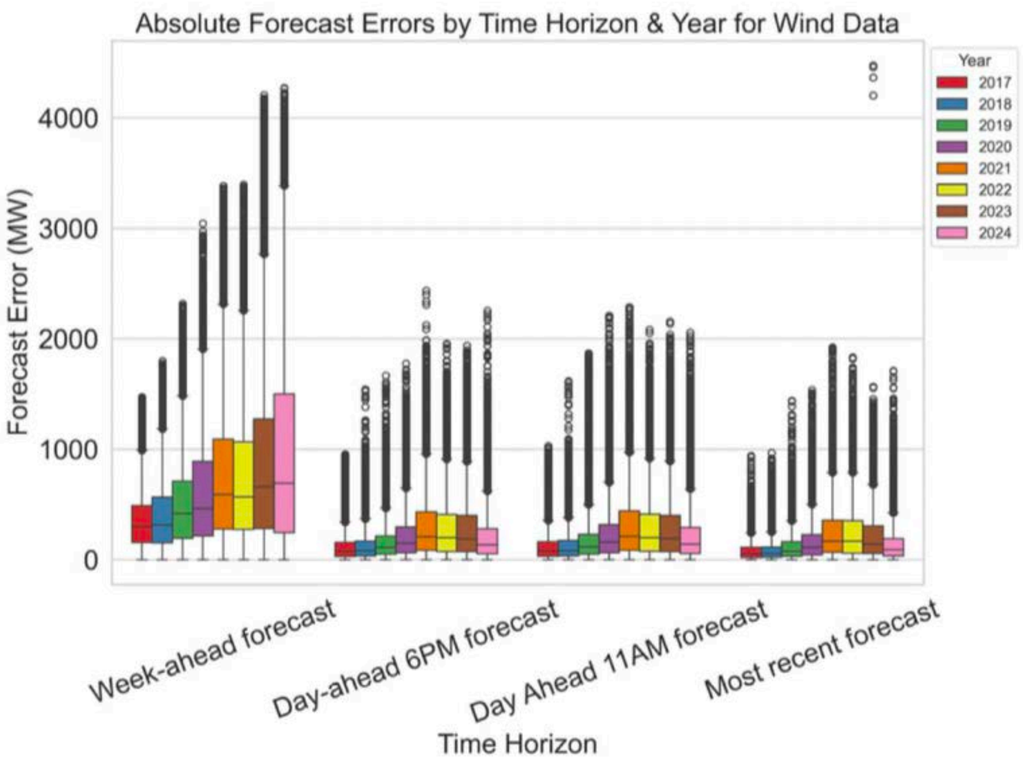
Week-ahead forecast

Most recent forecast

Elia. Open Data Platform.

Energy data and forecast quality

BE wind power generation forecast errors (2017 - 2024)

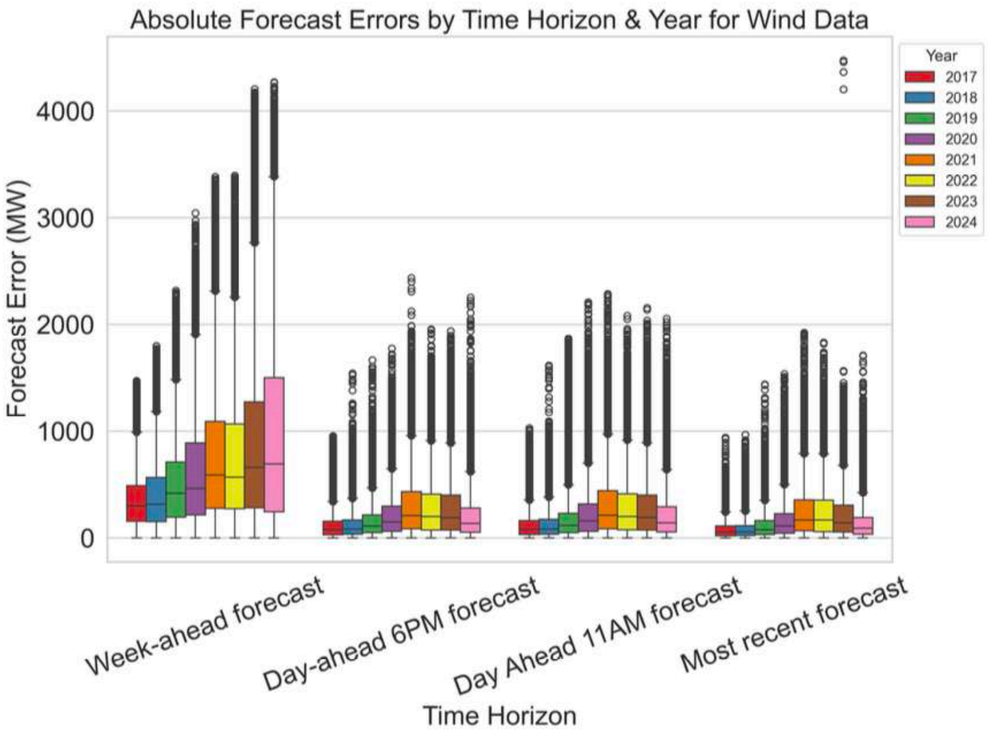


Absolute errors

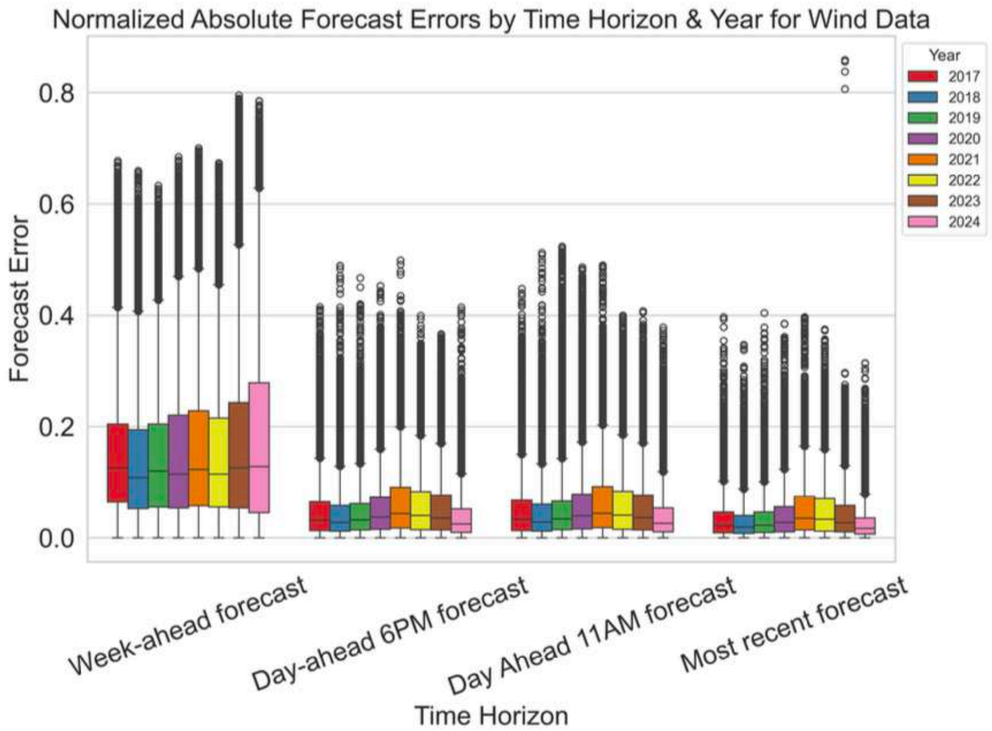
Canaydin, A., Uslu, Z. B., Depoortere, J., Mehmood, F., & Kazmi, H. (2025, November). Analysing and Improving National Renewable Energy Forecasts: A Case Study in Belgium. In *2025 IEEE PES Conference on Innovative Smart Grid Technologies-Middle East (ISGT Middle East)* (pp. 1-5). IEEE.

Energy data and forecast quality

BE wind power generation forecast errors (2017 - 2024)



Absolute errors



Capacity-normalized errors

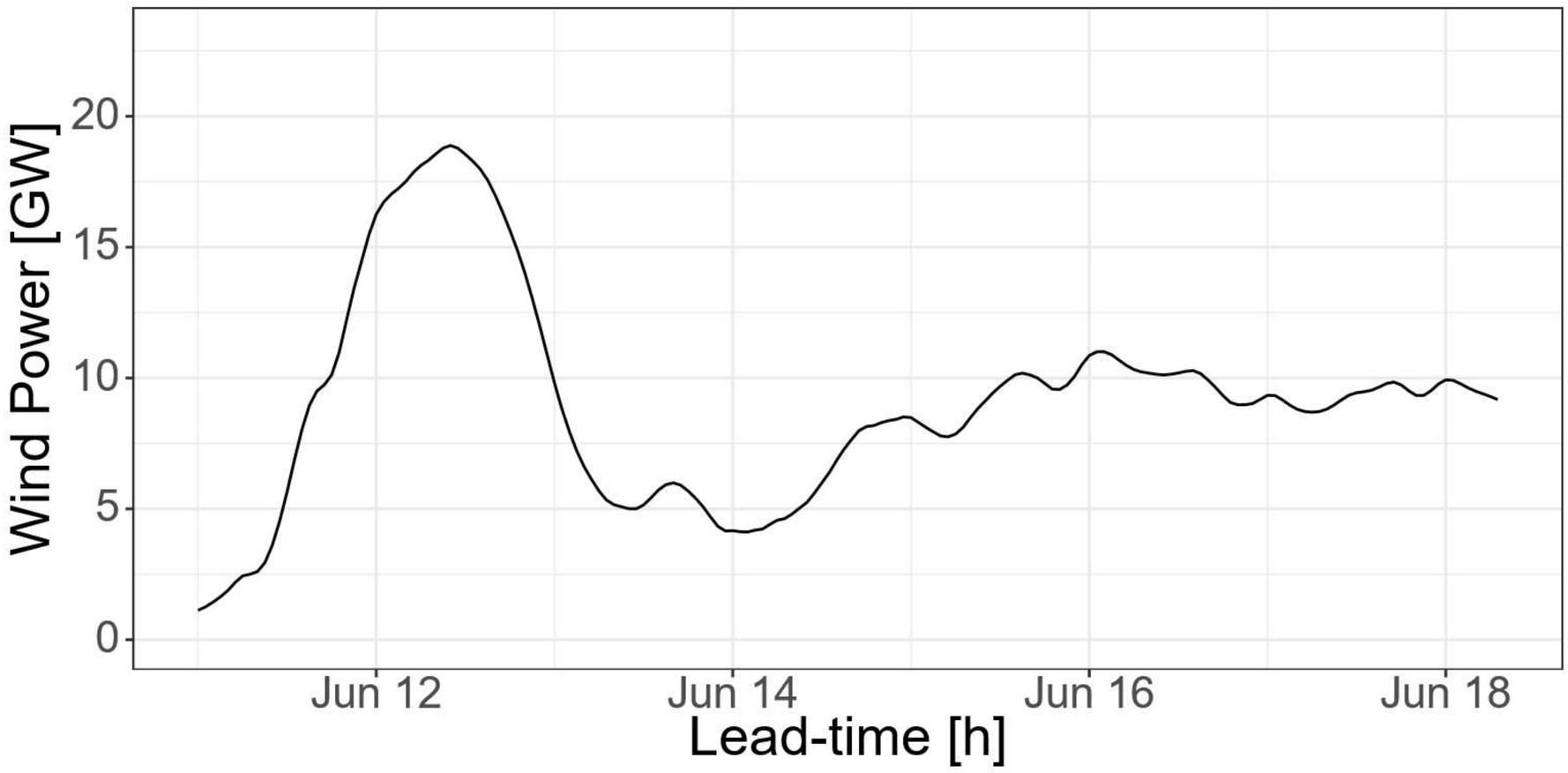
Canaydin, A., Uslu, Z. B., Depoortere, J., Mehmood, F., & Kazmi, H. (2025, November). Analysing and Improving National Renewable Energy Forecasts: A Case Study in Belgium. In *2025 IEEE PES Conference on Innovative Smart Grid Technologies-Middle East (ISGT Middle East)* (pp. 1-5). IEEE.

How do we address this challenge?

Possible solutions

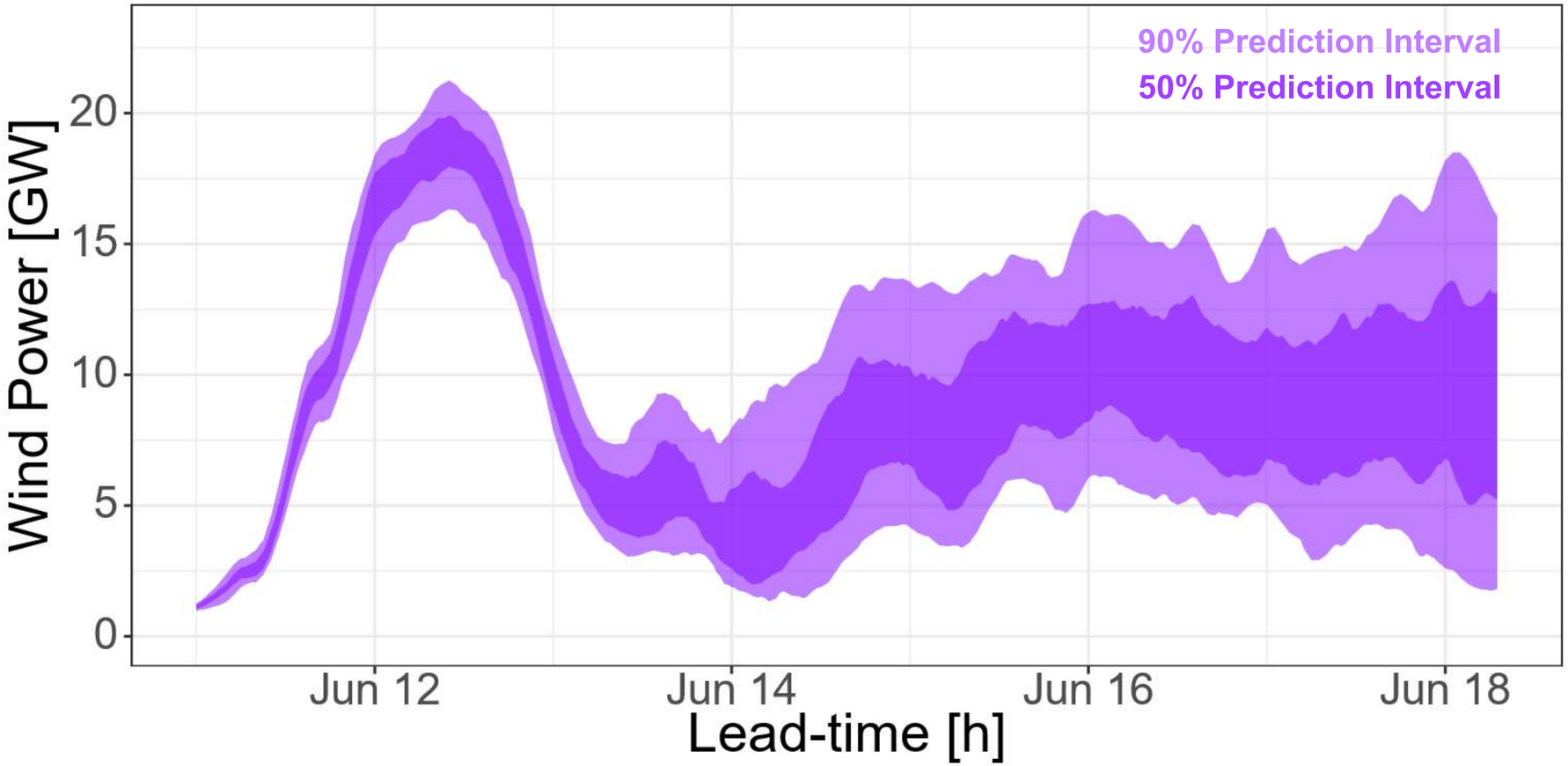
- Just make better forecasts (duh!)
- Incorporate uncertainty in your forecasts

Incorporating uncertainty in forecasts



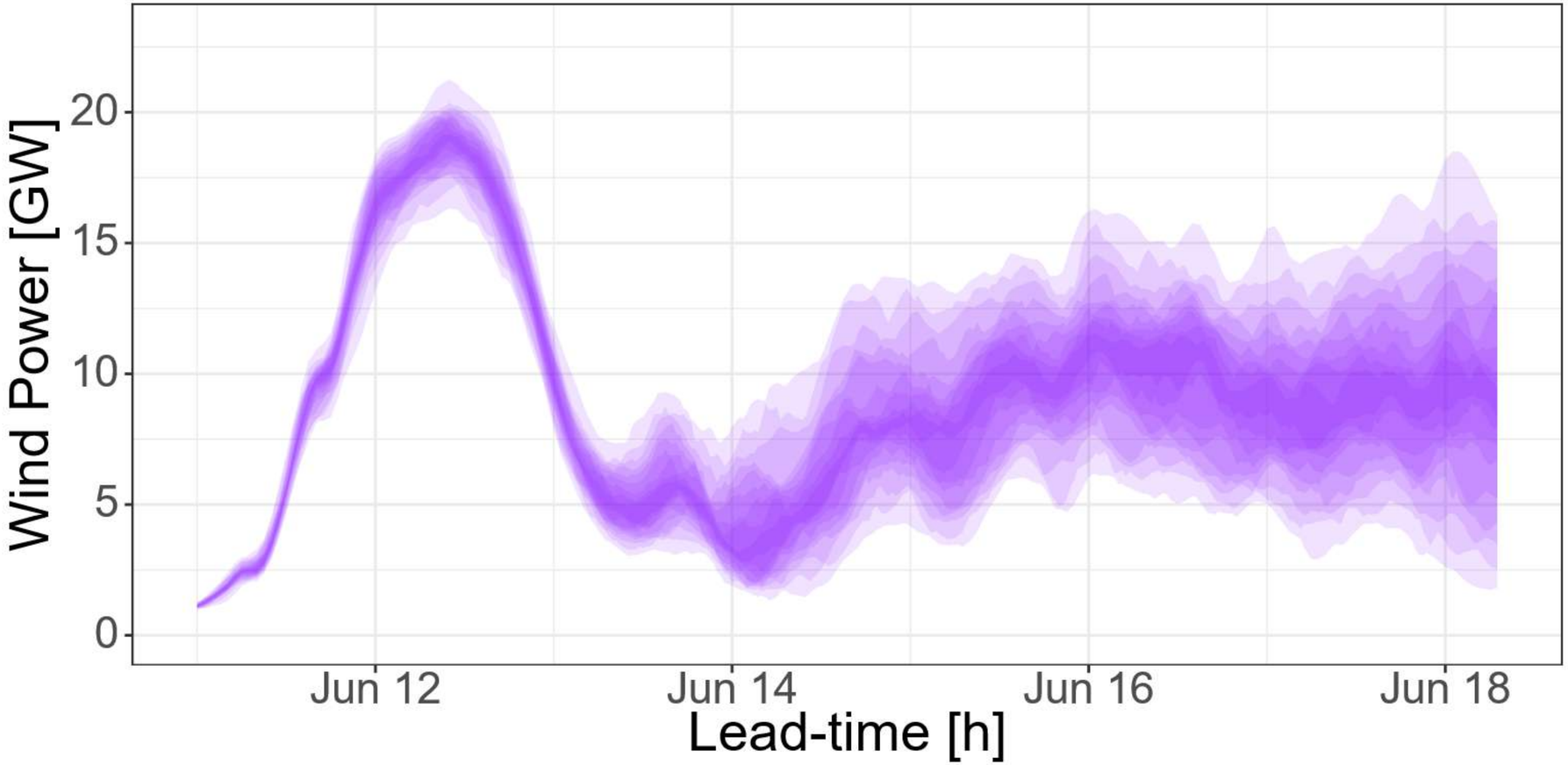
Jethro Browell (2025). Winning at wind power forecasting.

Incorporating uncertainty - prediction intervals



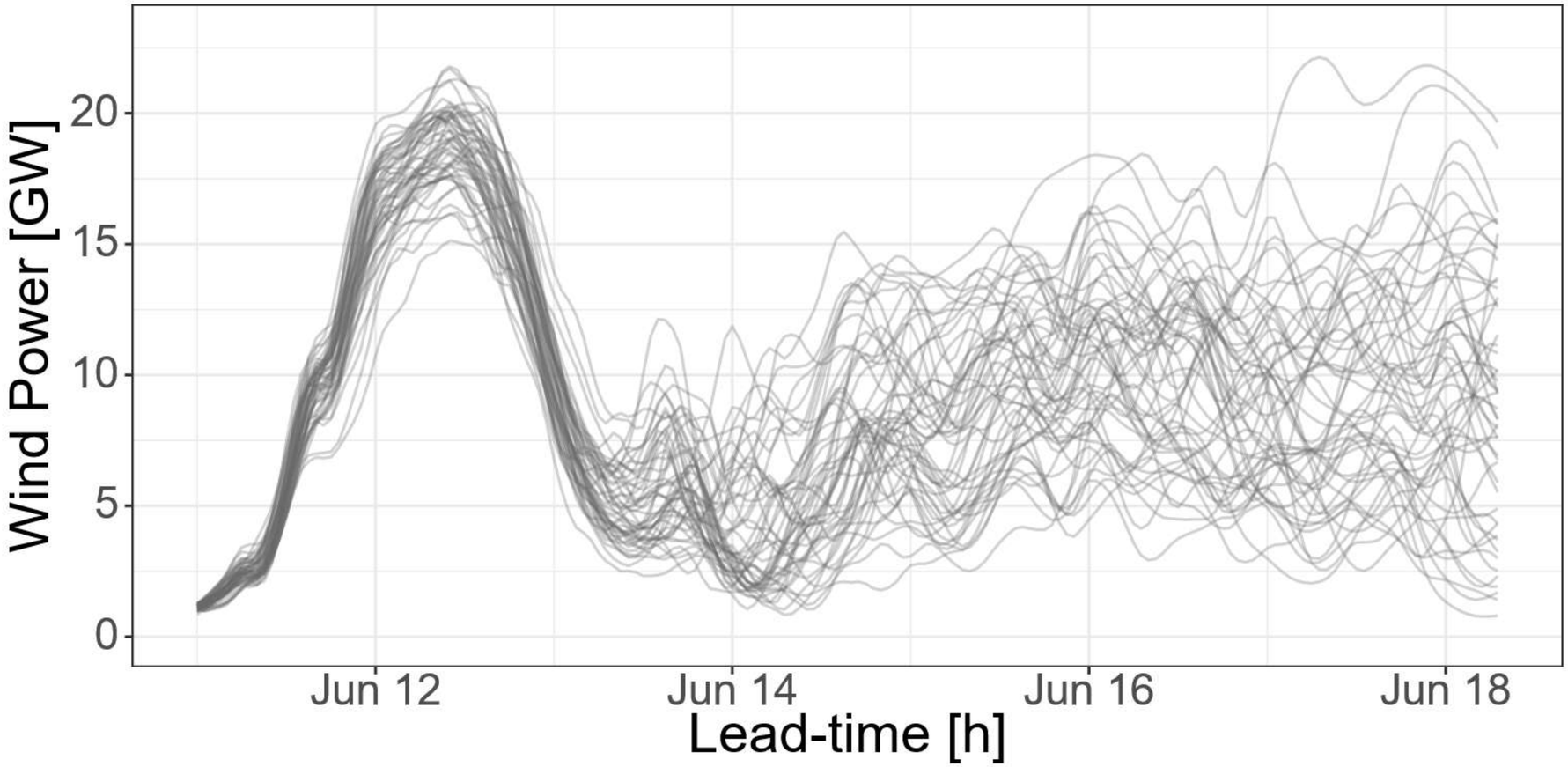
Jethro Browell (2025). Winning at wind power forecasting.

Incorporating uncertainty - distributions



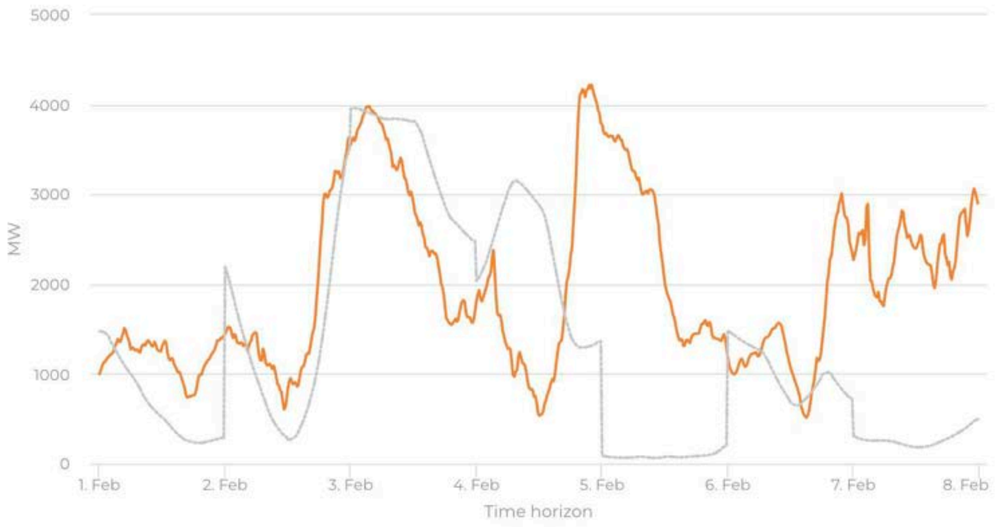
Jethro Browell (2025). Winning at wind power forecasting.

Incorporating uncertainty - trajectories

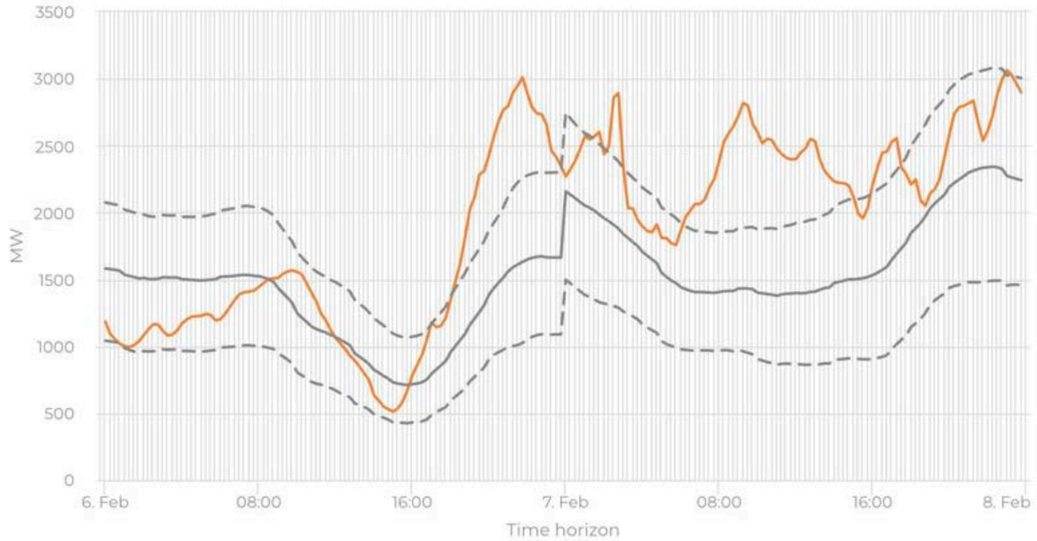


Jethro Browell (2025). Winning at wind power forecasting.

Some short-term examples (wind power - BE)



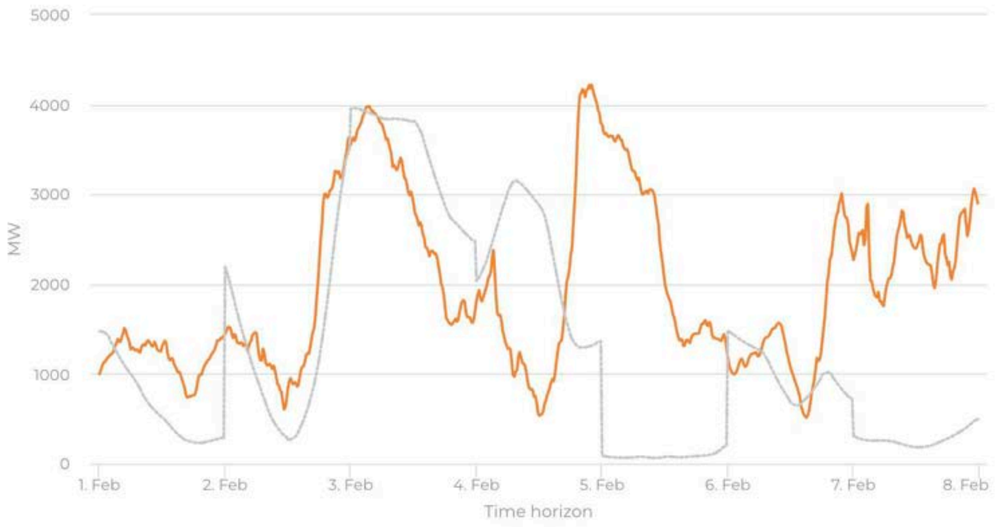
Week-ahead forecast



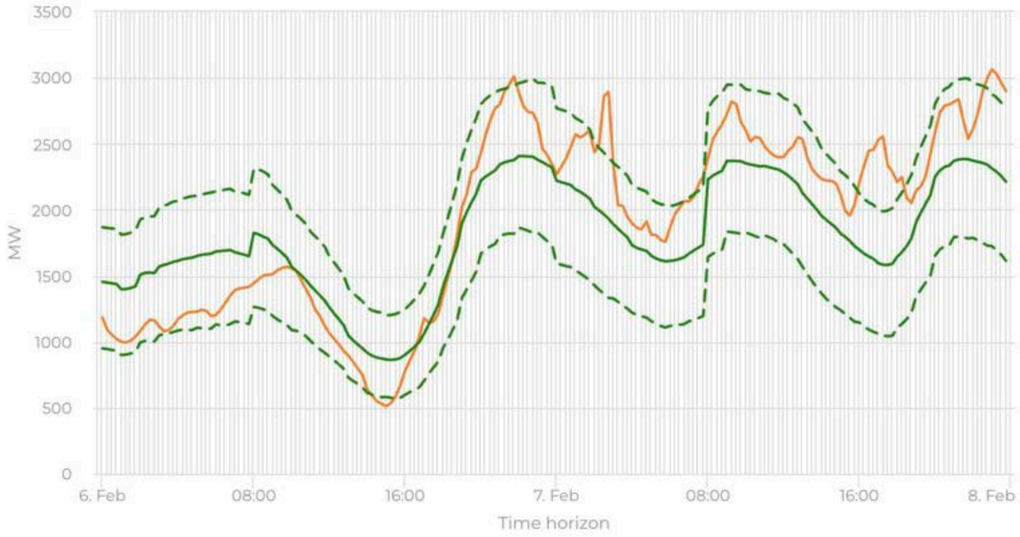
Day-ahead forecast

Elia. Open Data Platform.

Some short-term examples (wind power - BE)



Week-ahead forecast

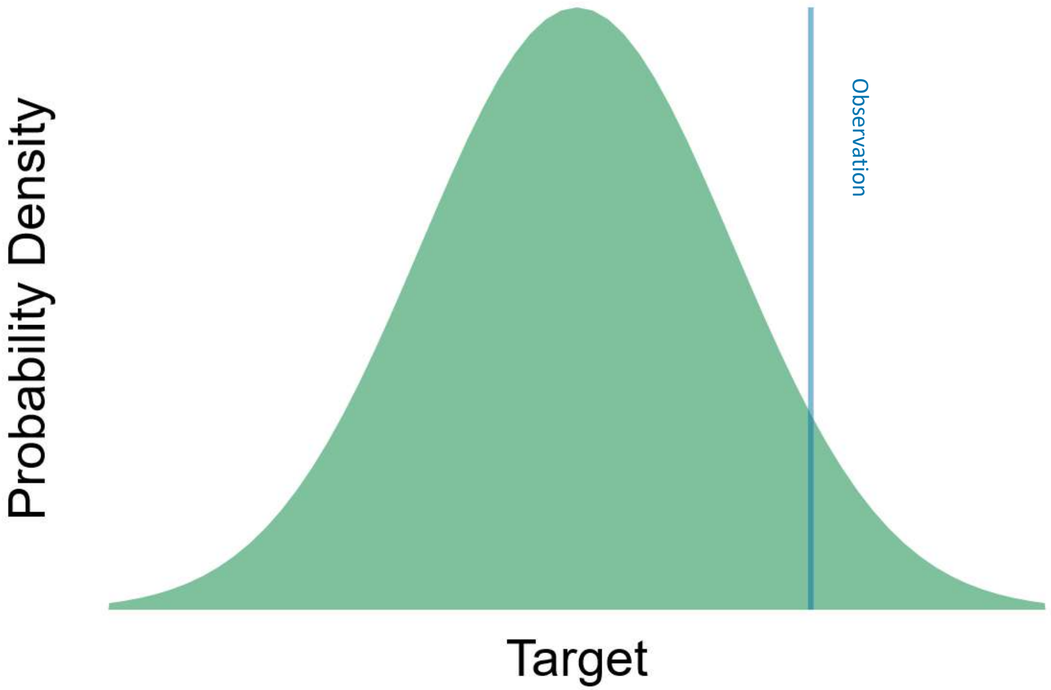


Most recent forecast

Elia. Open Data Platform.

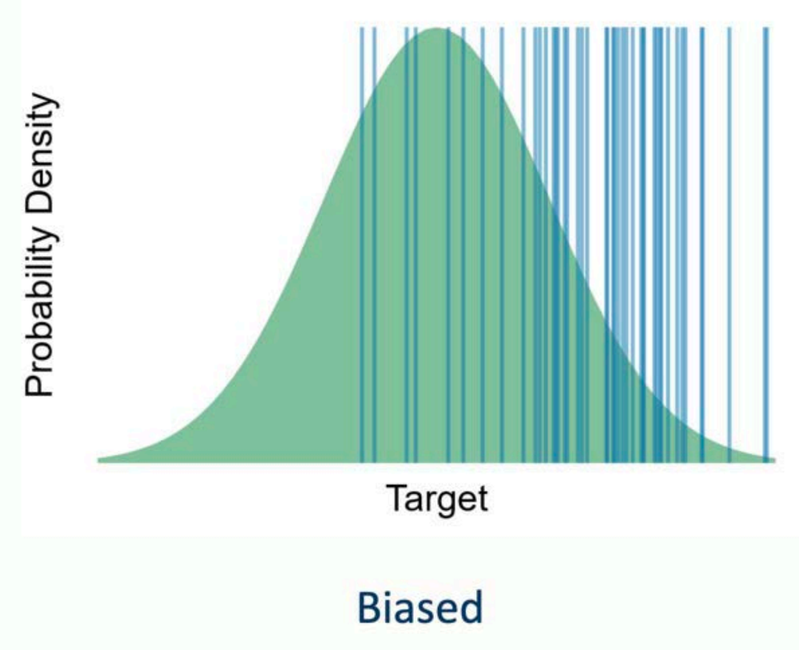
How do we evaluate a probabilistic forecast?

How do we evaluate a probabilistic forecast?



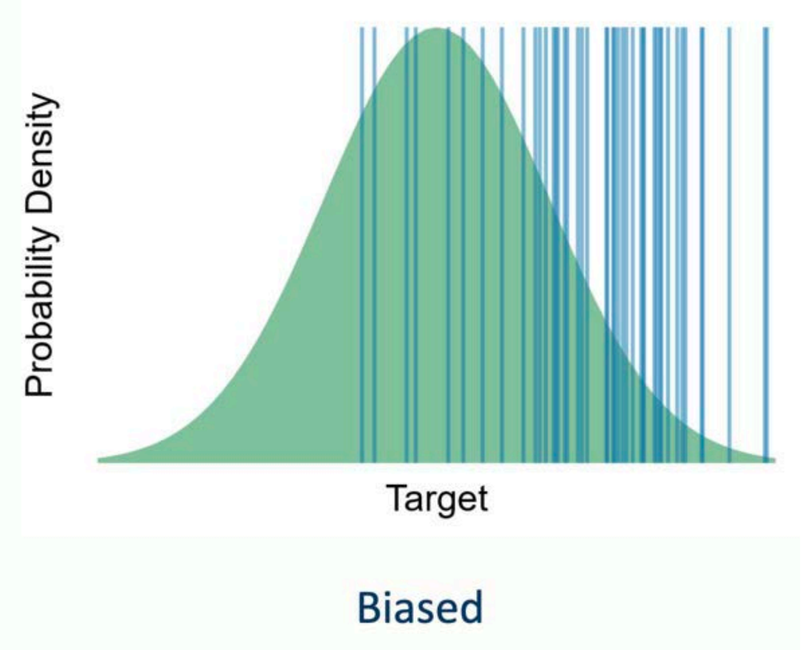
Jethro Browell (2025). Winning at wind power forecasting.

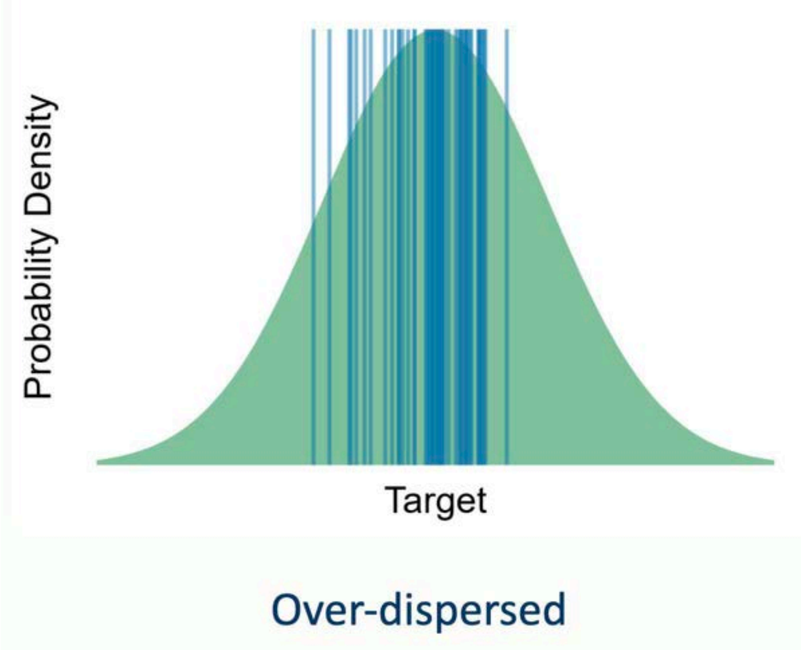
How do we evaluate a probabilistic forecast?



Jethro Browell (2025). Winning at wind power forecasting.

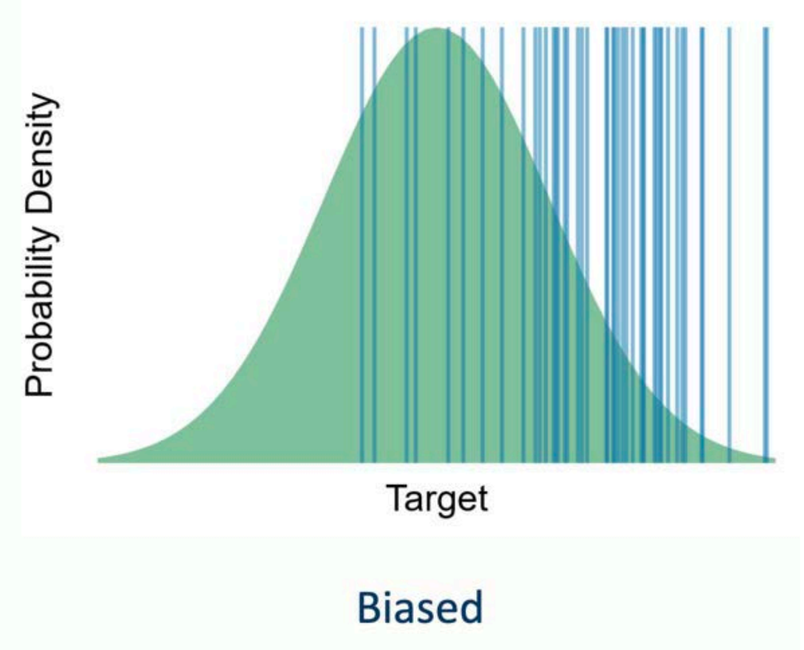
How do we evaluate a probabilistic forecast?

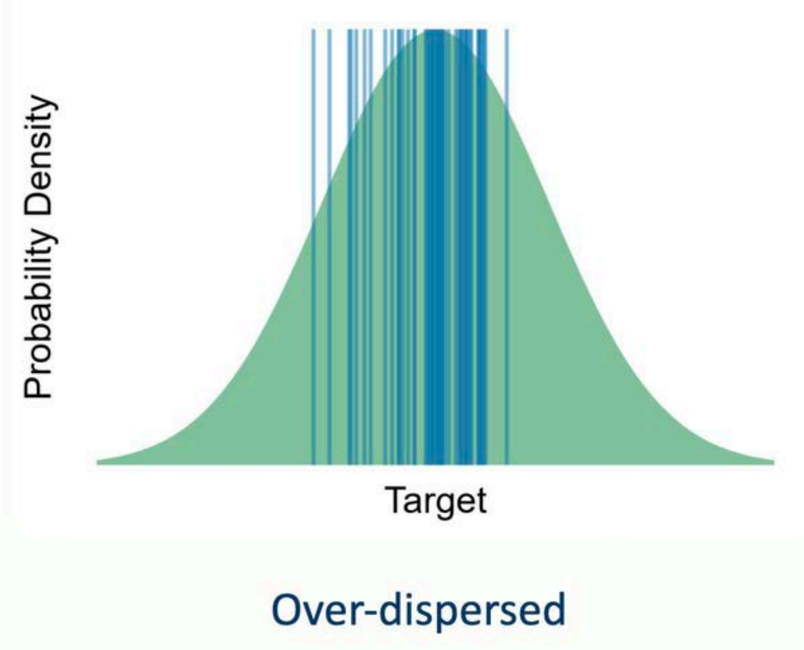


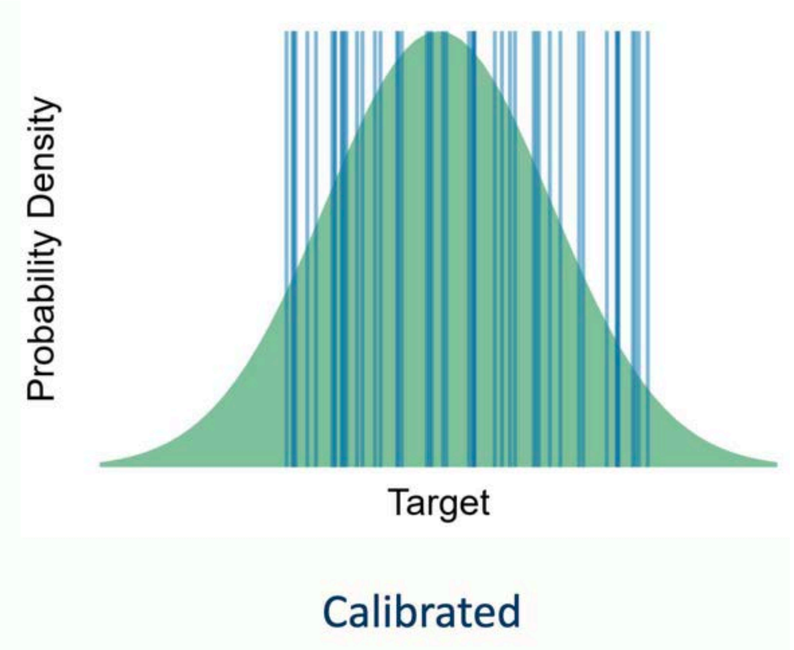


Jethro Browell (2025). Winning at wind power forecasting.

How do we evaluate a probabilistic forecast?







Jethro Browell (2025). Winning at wind power forecasting.

Is calibration enough on its own?

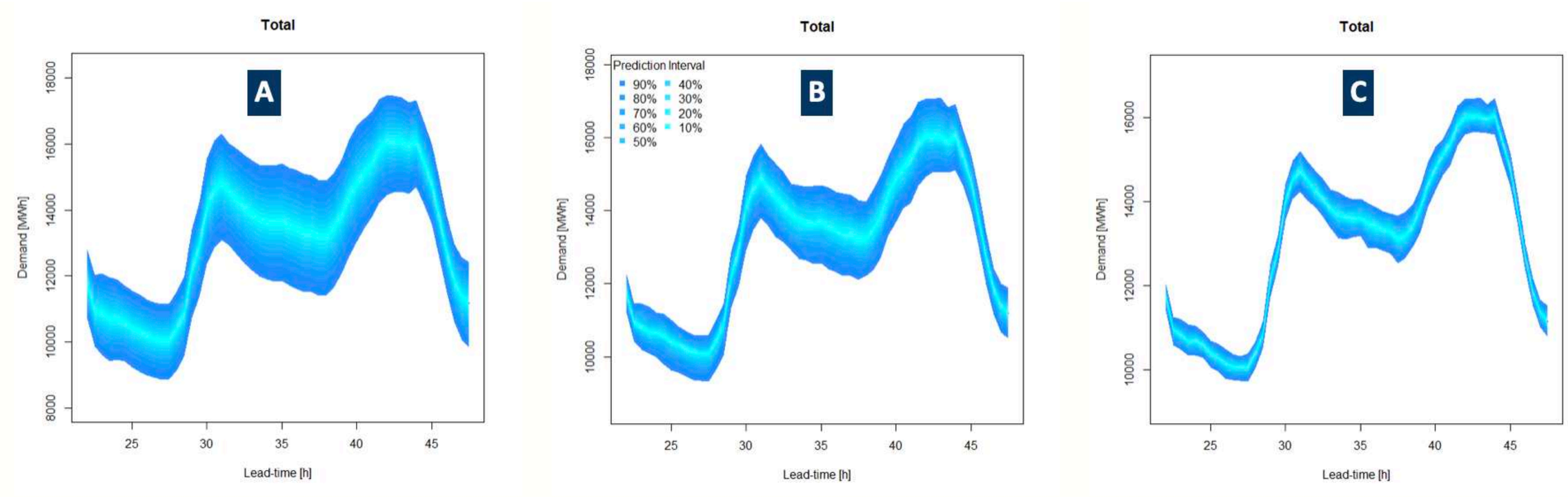
Some challenges with calibration

- Calibration is a summary statistic - no guarantee it will capture **all** the events with the same quality all the time (e.g. is the model calibrated on weekdays and weekends)
  - Check for conditional calibration

Some challenges with calibration

- Calibration is a summary statistic - no guarantee it will capture **all** the events with the same quality all the time (e.g. is the model calibrated on weekdays and weekends)
  - Check for conditional calibration
- For extreme intervals, predicting an infinitely wide interval may suffice to make a model well-calibrated
  - Check for sharpness

Sharpness



Jethro Browell (2025). Winning at wind power forecasting.

What makes a good probabilistic forecast?

- (Conditionally) well-calibrated and sharp
- CRPS / pinball etc.
- Other criterion
  - Stable
  - …

How to make a probabilistic forecast?

- Quantile regression

How to make a probabilistic forecast?

- Quantile regression
- Conformal prediction

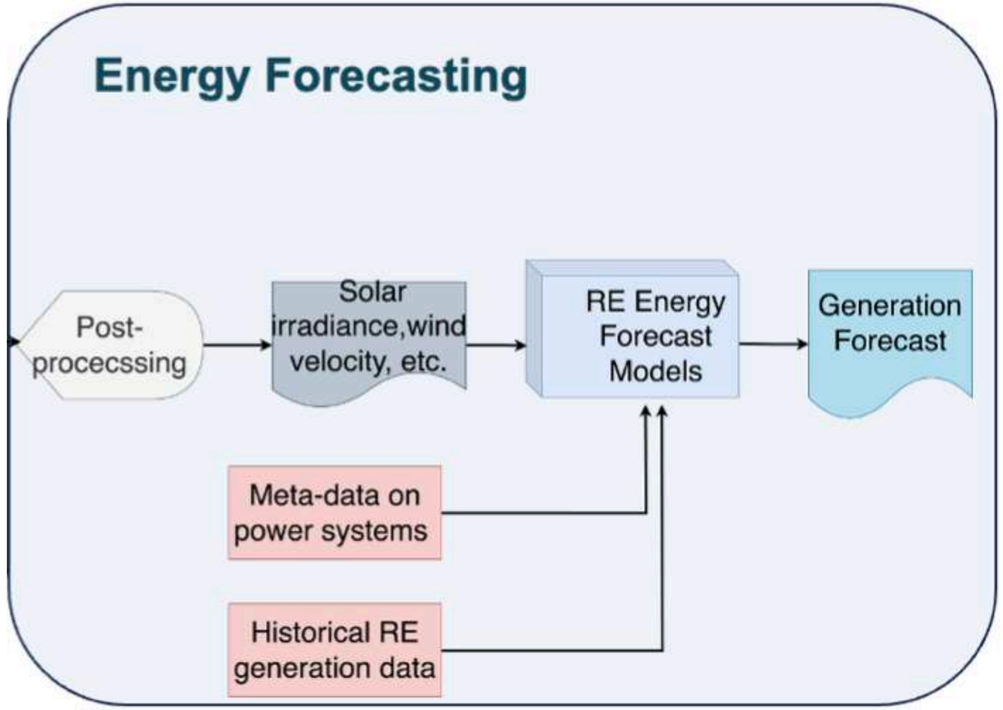
How to make a probabilistic forecast?

- Quantile regression
- Conformal prediction
- Ensemble to distribution forecasts
  - Ensemble of different models
  - Ensemble of same model with different inputs)
  - …

How do we make better energy forecasts?

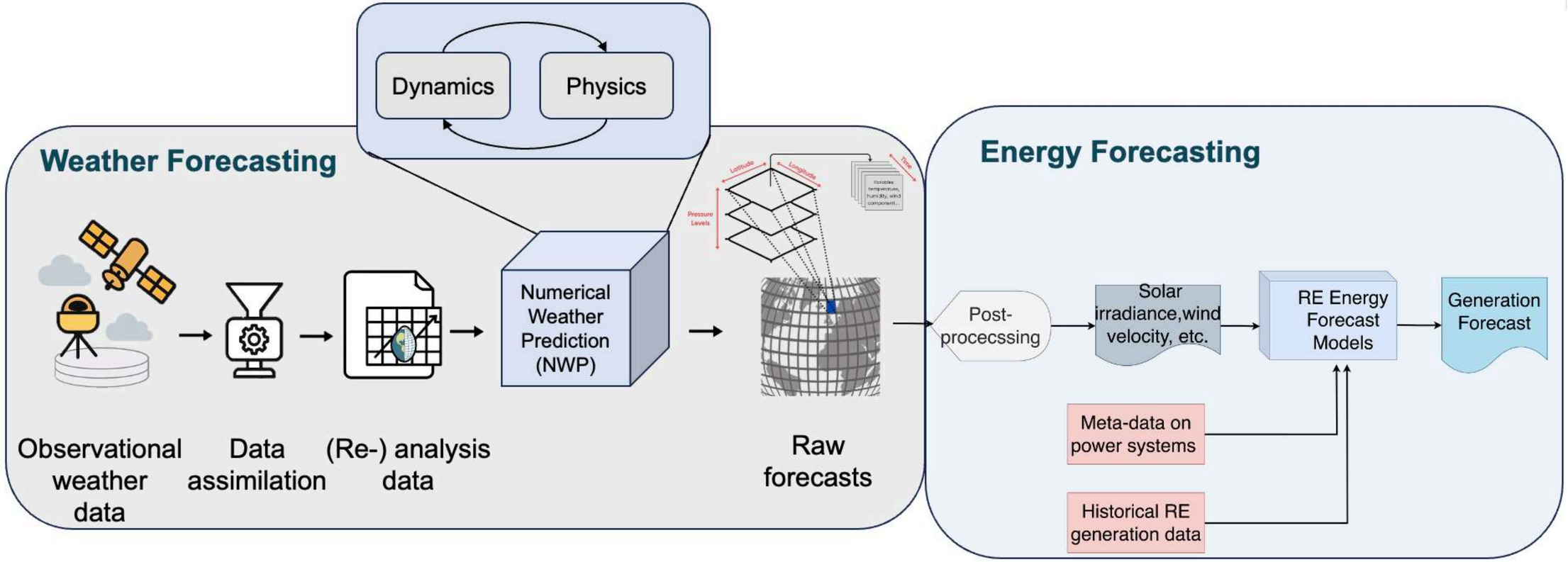
The energy forecasting workflow

From weather observations to renewable energy forecasts



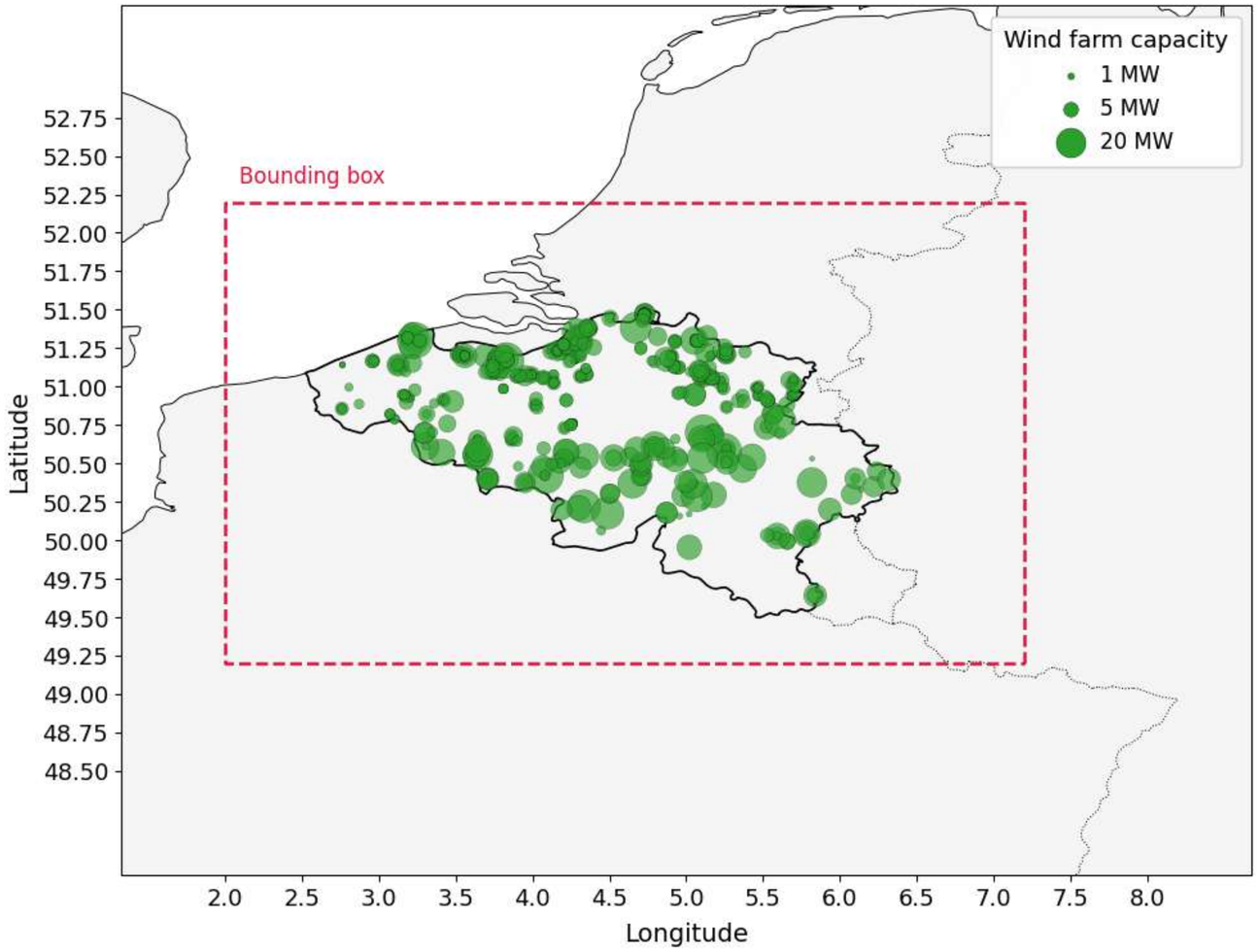
The energy forecasting workflow

From weather observations to renewable energy forecasts



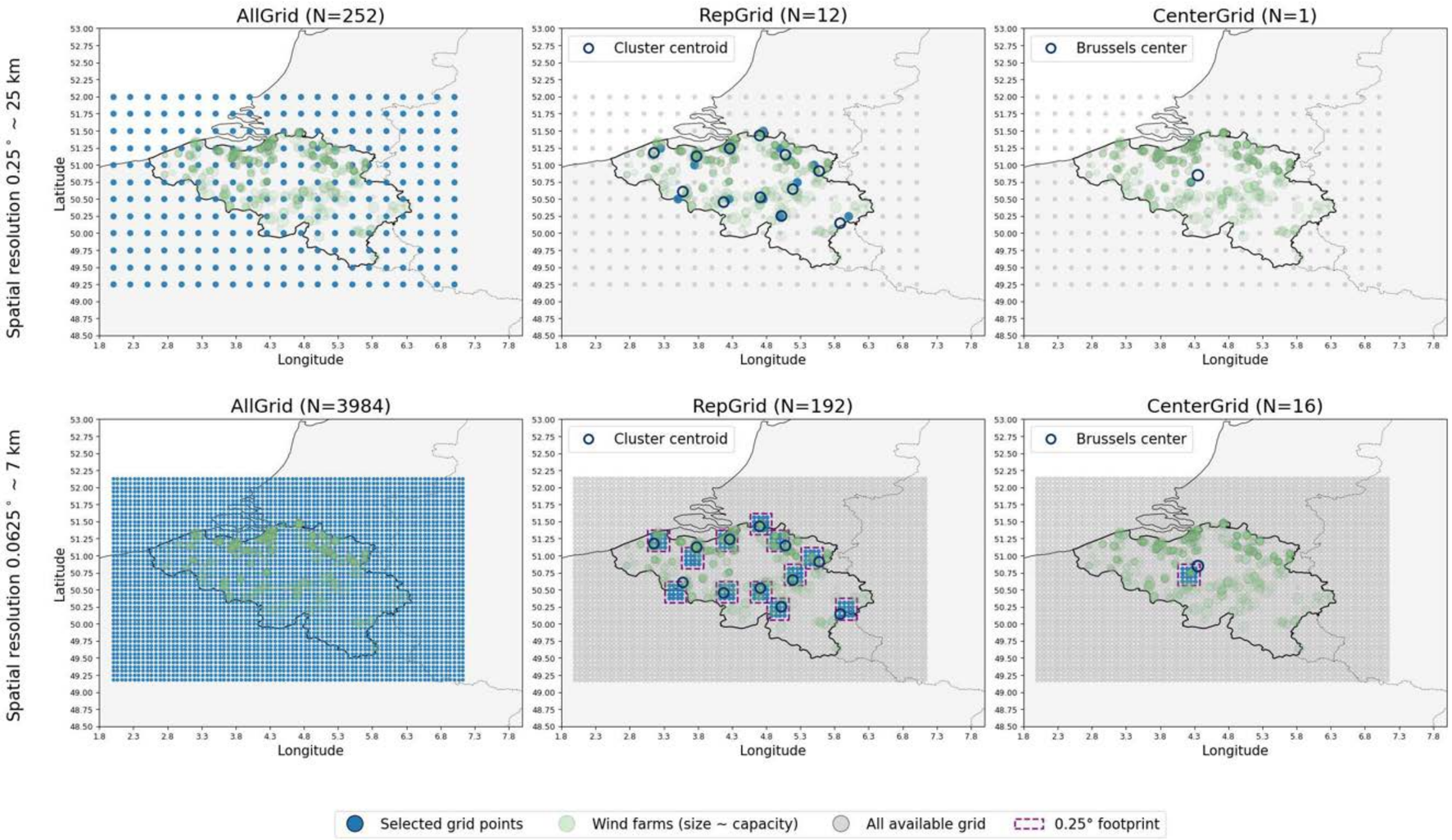
Forecasting onshore wind generation in BE

Experimental setup



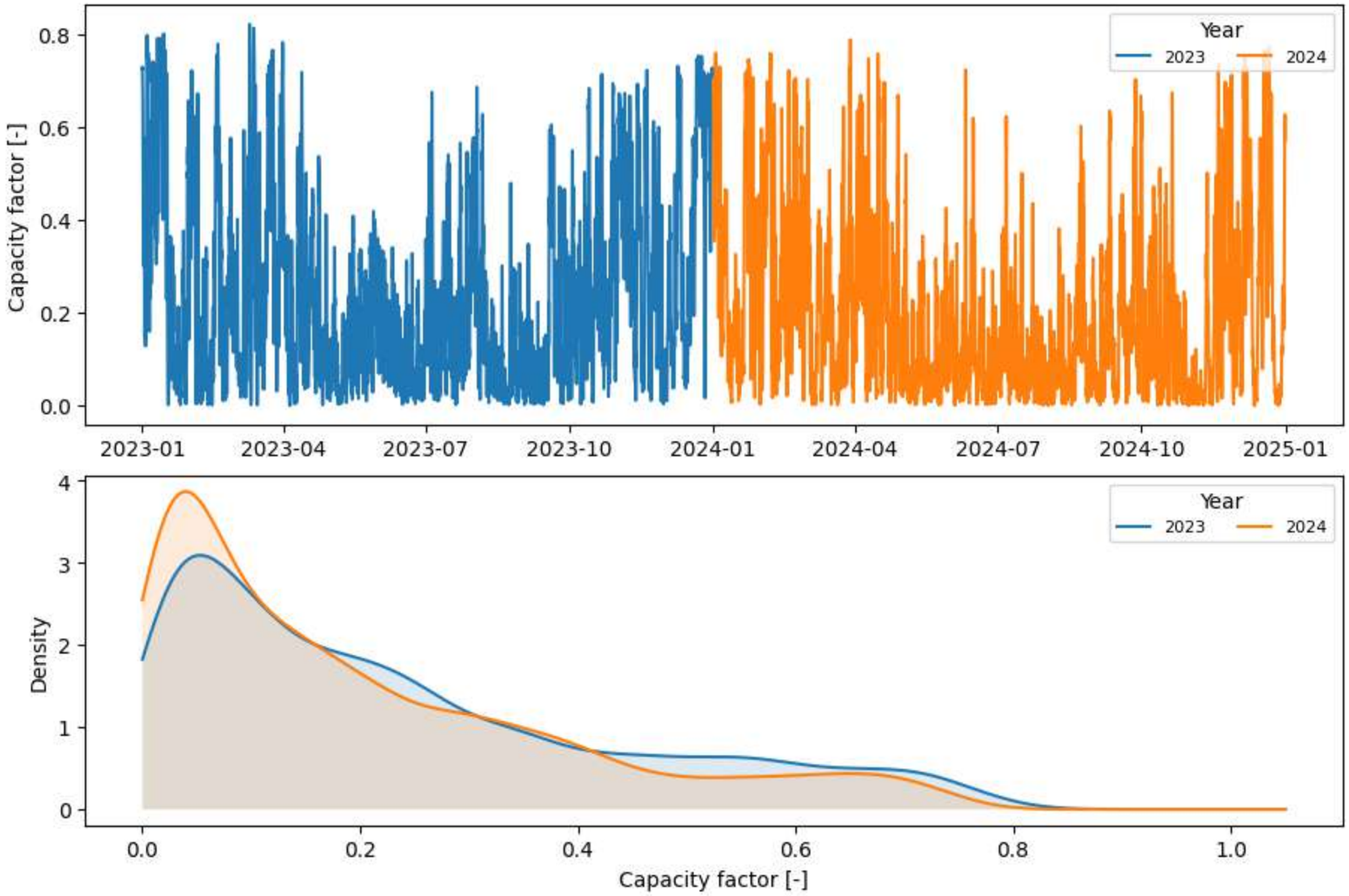
Forecasting onshore wind generation in BE

Which weather model and resolution to pick?



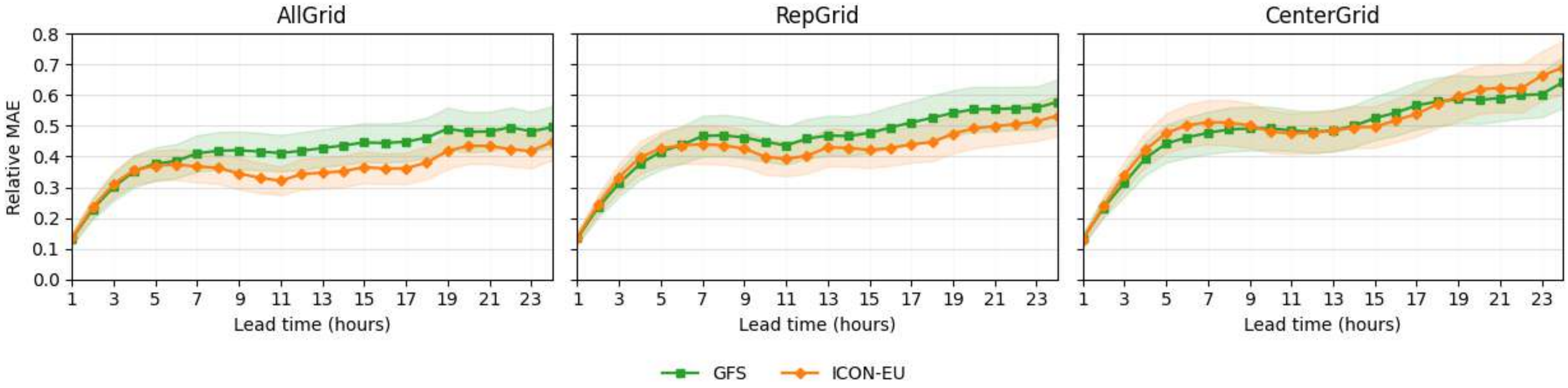
The energy forecasting problem

Experimental setup



The energy forecasting problem

Experimental setup



The energy forecasting problem

Experimental setup

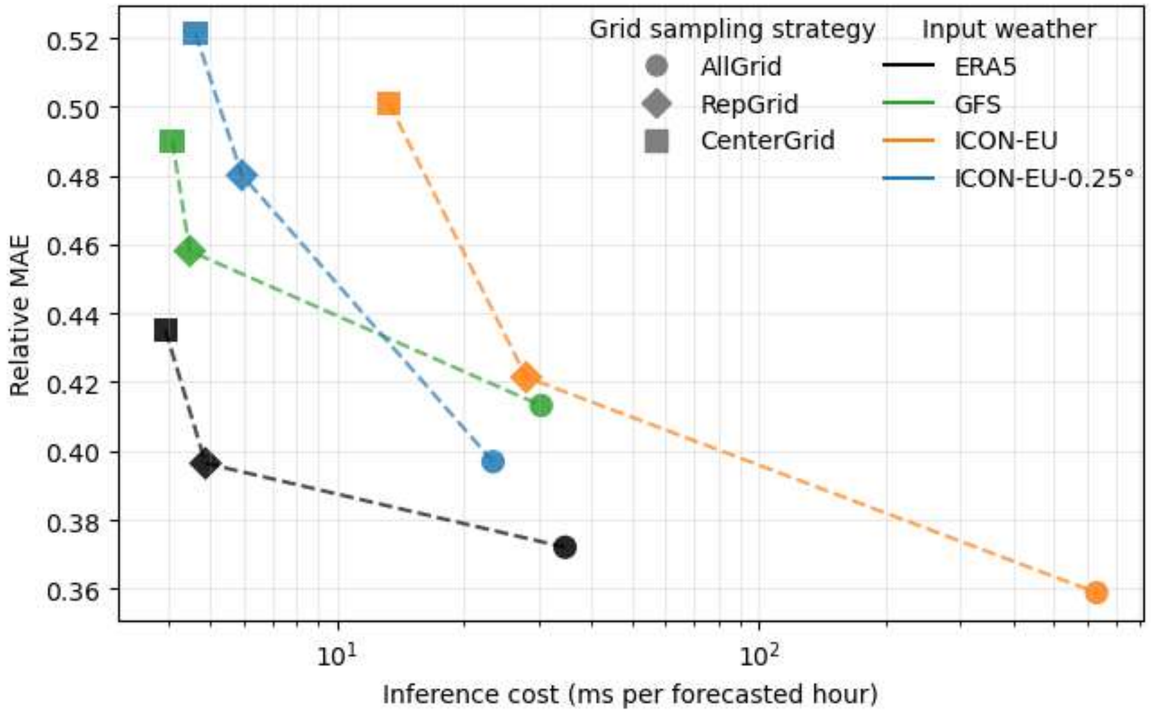
	RMSE								
	ERA5			GFS			ICON-EU		
Decile	AllGrid	RepGrid	CenterGrid	AllGrid	RepGrid	CenterGrid	AllGrid	RepGrid	CenterGrid
1	0.061	0.072	0.078	0.064	0.086	0.091	**0.051**	0.071	0.097
2	0.068	0.077	0.082	0.075	0.088	0.094	**0.060**	0.075	0.097
3	0.060	0.062	0.066	0.063	0.070	0.075	**0.055**	0.063	0.080
4	0.059	0.059	0.064	0.066	0.065	0.073	**0.056**	0.062	0.074
5	0.058	0.059	0.065	0.067	0.071	0.077	**0.054**	0.063	0.078
6	0.052	0.053	0.058	0.061	0.062	0.068	**0.051**	0.058	0.071
7	**0.057**	0.062	0.065	0.066	0.069	0.073	0.062	0.066	0.079
8	**0.066**	0.076	0.081	0.074	0.085	0.092	0.069	0.081	0.096
9	0.094	0.105	0.115	0.109	0.121	0.132	**0.091**	0.108	0.127
10	0.151	0.150	0.161	0.164	0.164	0.171	**0.139**	0.152	0.168
Overall	0.078	0.082	0.089	0.087	0.093	0.100	**0.074**	0.085	0.101

The ICON-EU outperforms the lower resolution GFS in all cases

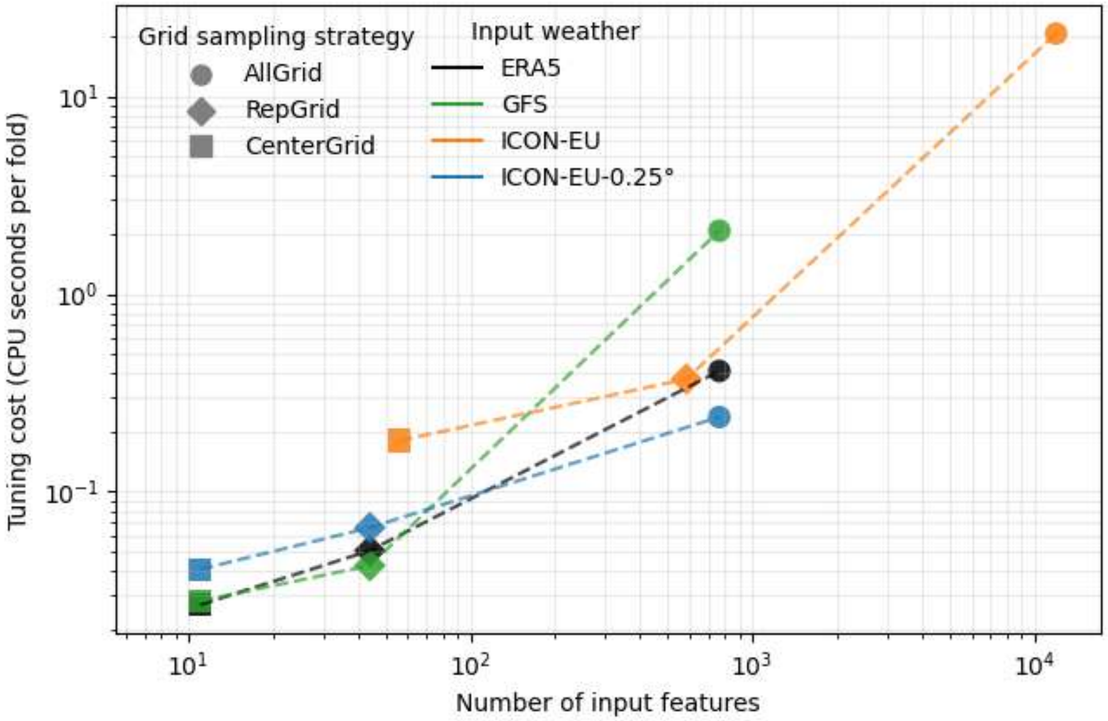
AllGrid setups also outperform lower grid point setups by a large margin

The energy forecasting problem

Experimental setup



Compute vs. error



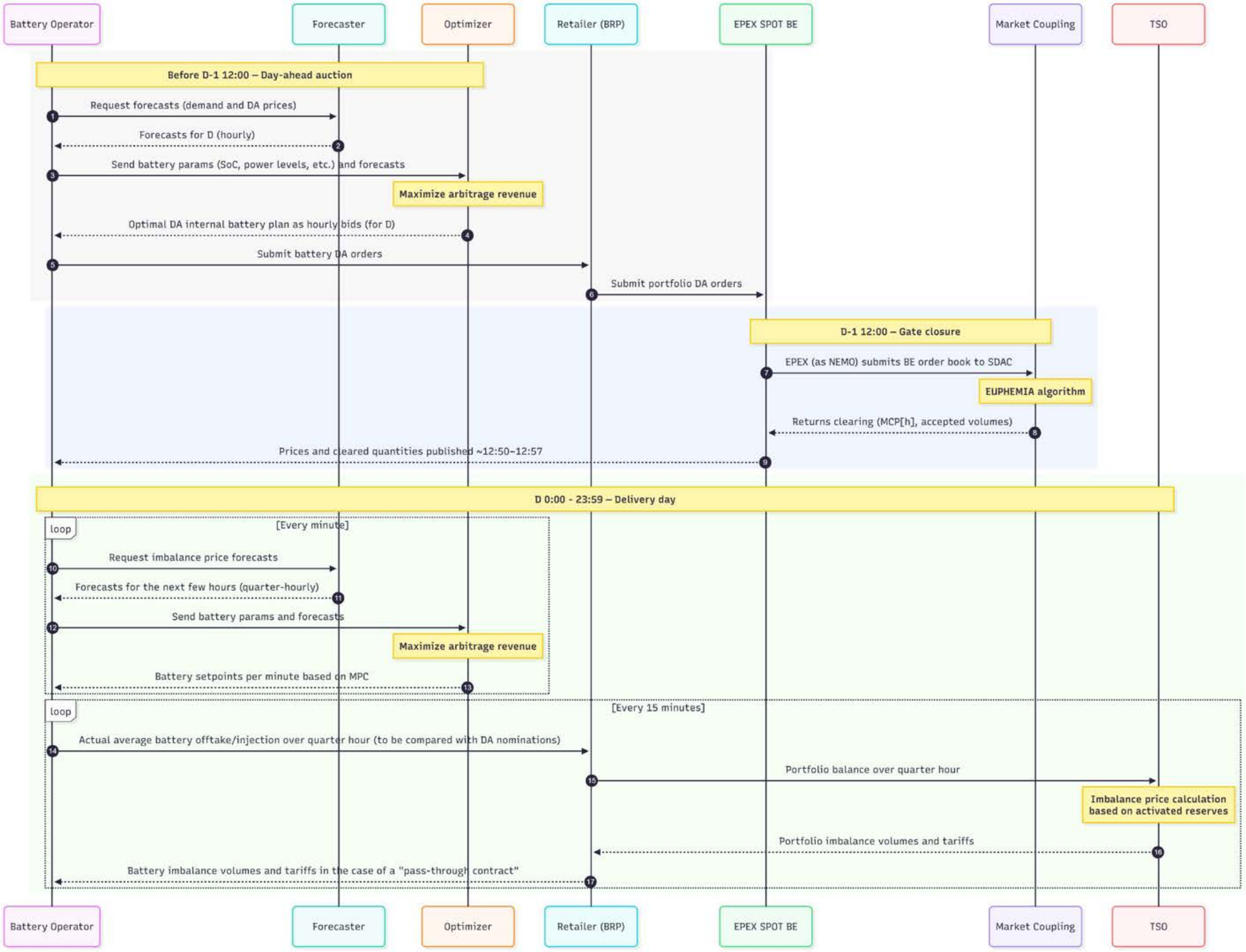
Input features vs. compute

Concluding remarks

The long road ahead

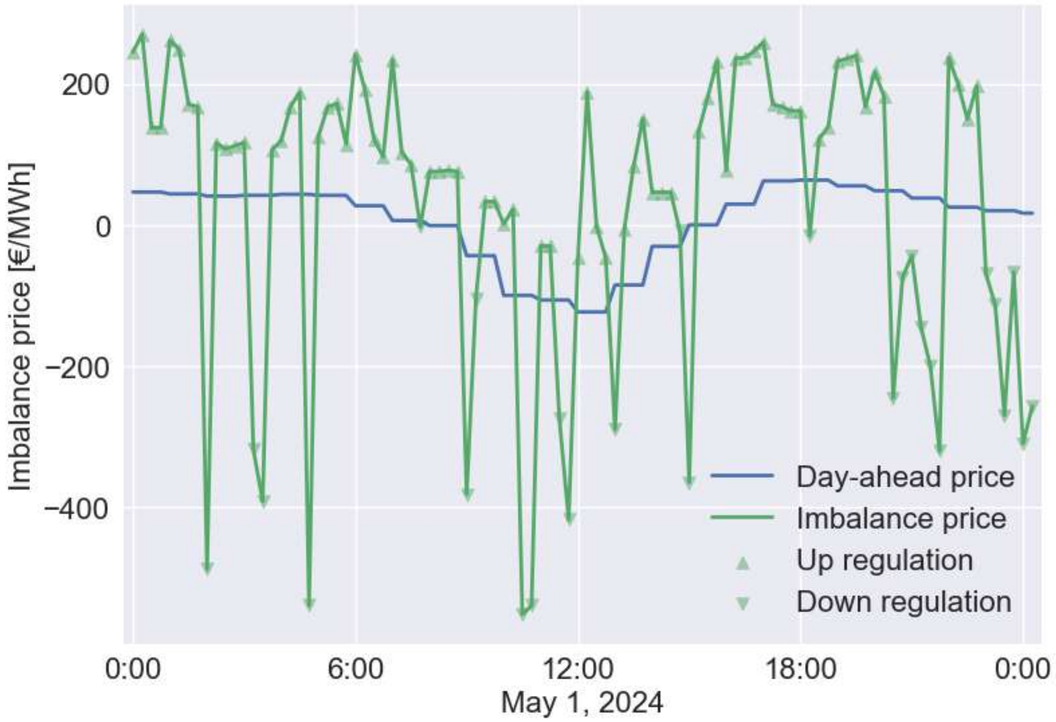
- Forecasting and scheduling renewables is an increasingly challenging problem, with grave implications for security of supply
  - Upstream weather forecast uncertainty is increasingly driving generation uncertainty in Europe
  - Generation uncertainty is coupled with market uncertainty, and detracts from cost-optimal scheduling of available resources
- Better observability, data and forecast algorithms (incl. in quantifying and managing uncertainty) holds the promise to vastly improve the current status quo

The role of market forecasts in energy forecasts


Battery Operator
Forecaster
Optimizer
Retailer (BRP)
EPEX SPOT BE
Market Coupling
TSO
Before D-1 12:00 – Day-ahead auction
1 Request forecasts (demand and DA prices)
2 Forecasts for D (hourly)
3 Send battery params (SoC, power levels, etc.) and forecasts
Maximize arbitrage revenue
4 Optimal DA internal battery plan as hourly bids (for D)
5 Submit battery DA orders
6 Submit portfolio DA orders
D-1 12:00 – Gate closure
7 EPEX (as NEMO) submits BE order book to SDAC
EUPHEMIA algorithm
8 Returns clearing (MCP[h], accepted volumes)
9 Prices and cleared quantities published ~12:50–12:57
D 0:00 - 23:59 – Delivery day
loop
[Every minute]
10 Request imbalance price forecasts
11 Forecasts for the next few hours (quarter-hourly)
12 Send battery params and forecasts
Maximize arbitrage revenue
13 Battery setpoints per minute based on MPC
loop
[Every 15 minutes]
14 Actual average battery offtake/injection over quarter hour (to be compared with DA nominations)
15 Portfolio balance over quarter hour
Imbalance price calculation based on activated reserves
16 Portfolio imbalance volumes and tariffs
17 Battery imbalance volumes and tariffs in the case of a "pass-through contract"
Battery Operator
Forecaster
Optimizer
Retailer (BRP)
EPEX SPOT BE
Market Coupling
TSO

Understanding DAM and IMB prices in BE

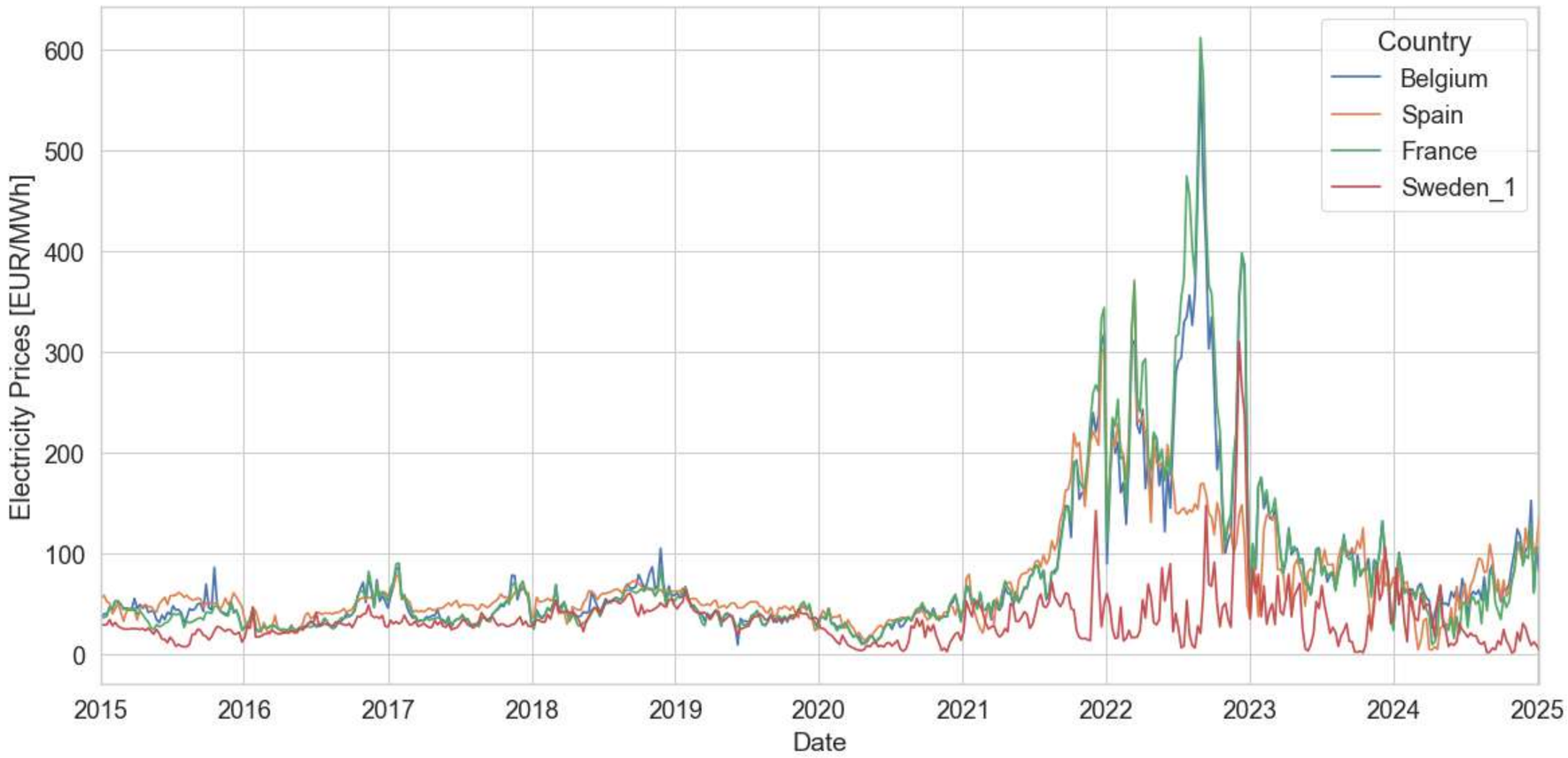
Increasing volatility on the IMB market



How well can we actually forecast electricity market prices?

Understanding DAM prices in BE

A decade of price volatility in four countries



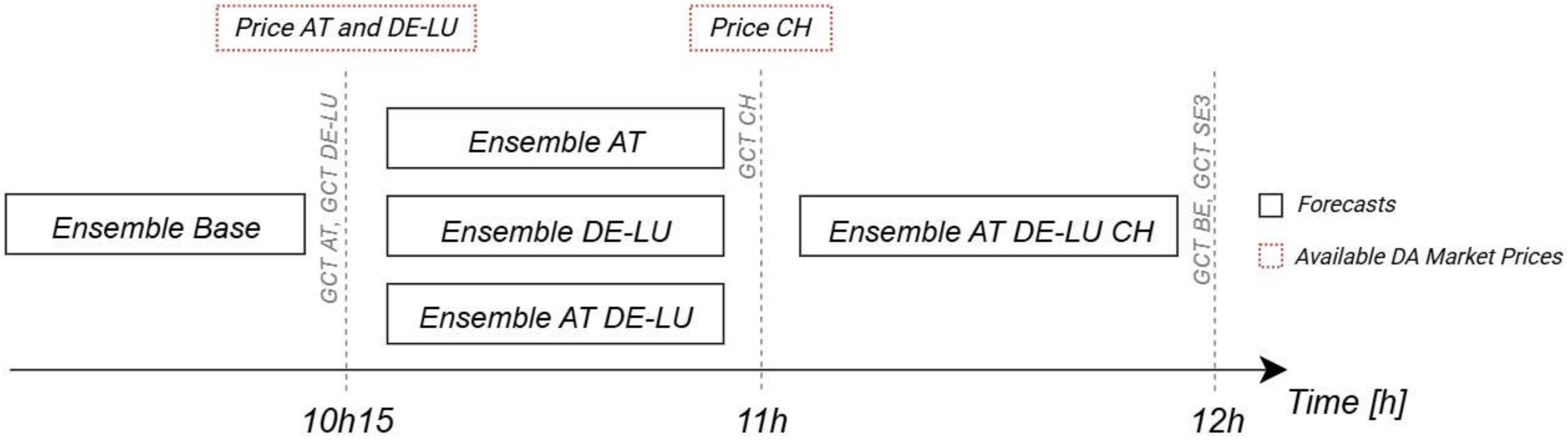
Forecasting the DAM prices in BE and SE(3)

The role of market information

Fundamental Drivers BE	Unit	Frequency	Source
Day-Ahead Load Forecast	MW	Hourly	Entso-E
Wind Forecast (Onshore & Offshore)	MW	Hourly	Entso-E
Solar Forecast	MW	Hourly	Entso-E
Temperature Forecast	°C	Hourly	Open-meteo
Humidity Forecast	%	Hourly	Open-meteo
Day-of-Week Indicator	–	Daily	National Calendar
Public Holiday Indicator	Binary	Daily	National Calendar
Fundamental Drivers SE3	Unit	Frequency	Source
Day-Ahead Load Forecast	MW	Hourly	Entso-E
Wind Forecast (Onshore)	MW	Hourly	Entso-E
Temperature Forecast	°C	Hourly	Open-meteo
Humidity Forecast	%	Hourly	Open-meteo
Day-of-Week Indicator	–	Daily	National Calendar
Public Holiday Indicator	Binary	Daily	National Calendar
Market Price Data	Unit	Frequency	Source
DE-LU Spot Price	€/MWh	15-min	Entso-E
AT Spot Price	€/MWh	15-min	Entso-E
CH Spot Price	€/MWh	Hourly	Entso-E

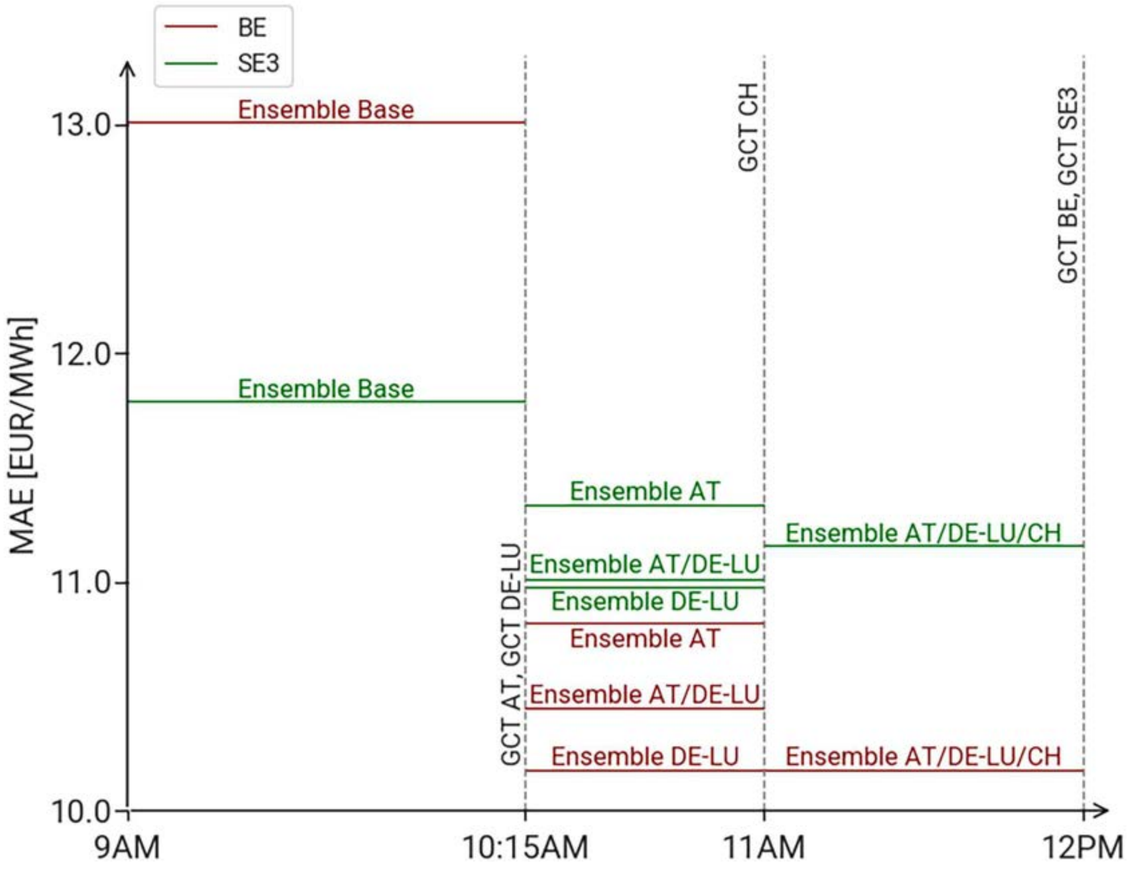
Forecasting the DAM prices in BE and SE(3)

The role of market information



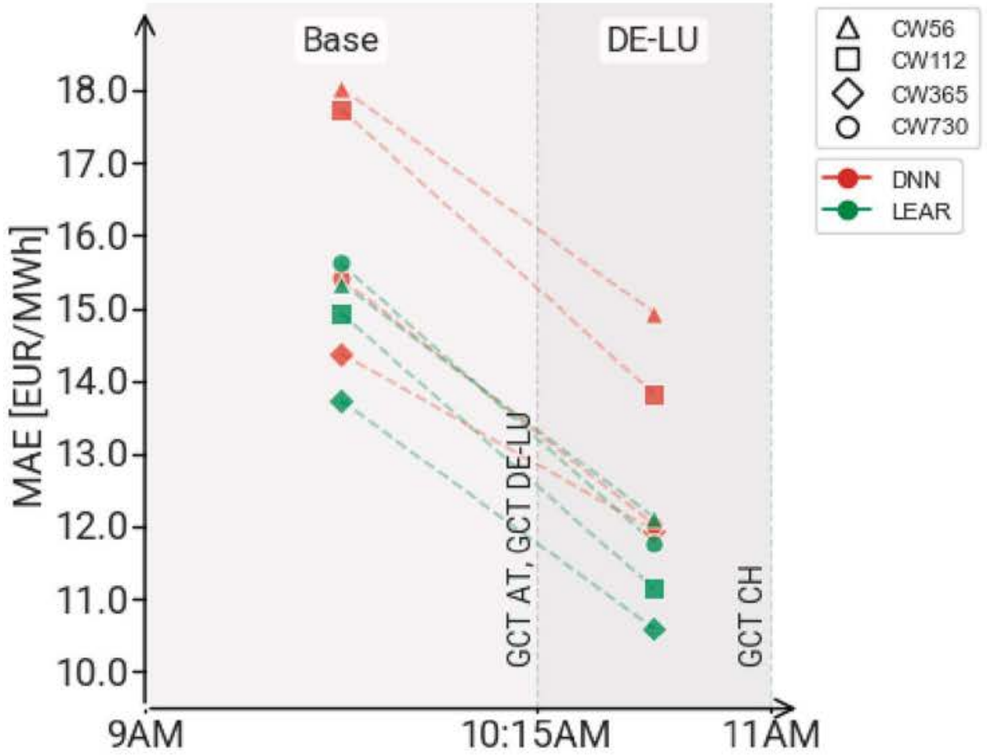
Forecasting the DAM prices in BE and SE(3)

The role of market information

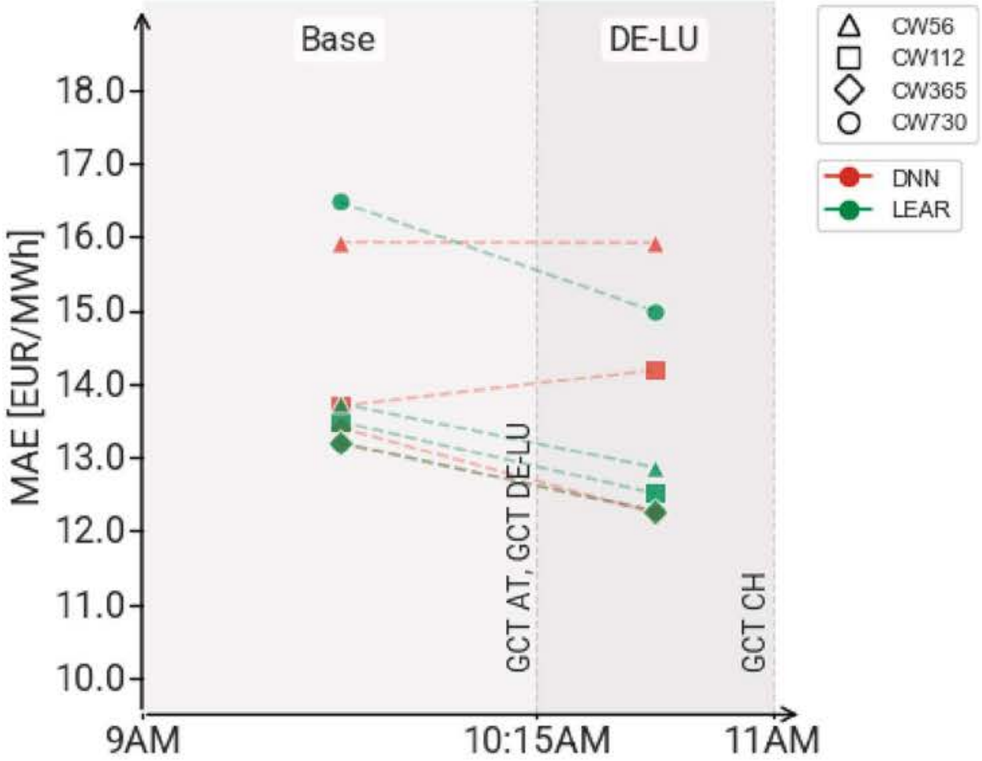


Forecasting the DAM prices in BE and SE(3)

The role of modeling choices (architecture, history, covariates …)



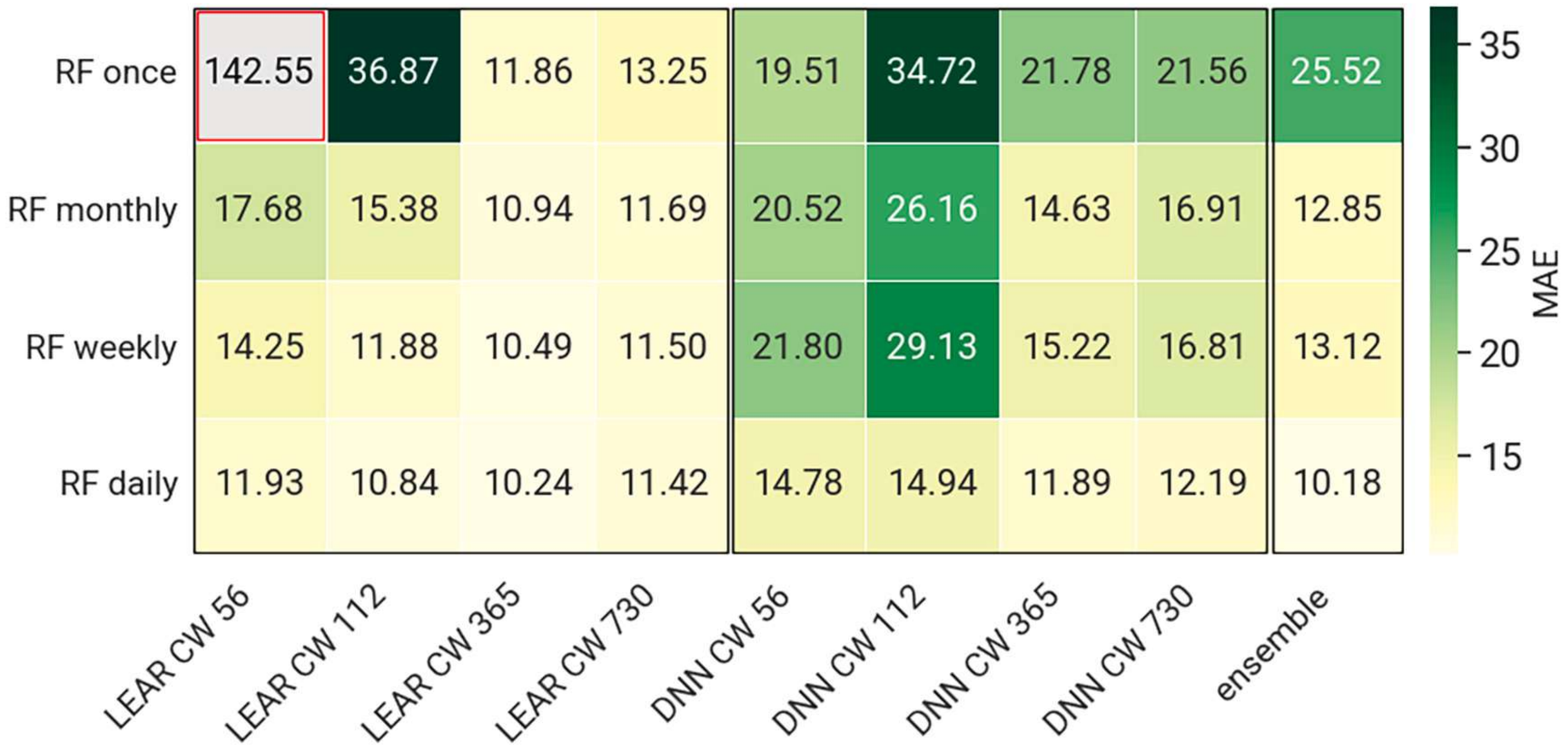
(a)



(b)

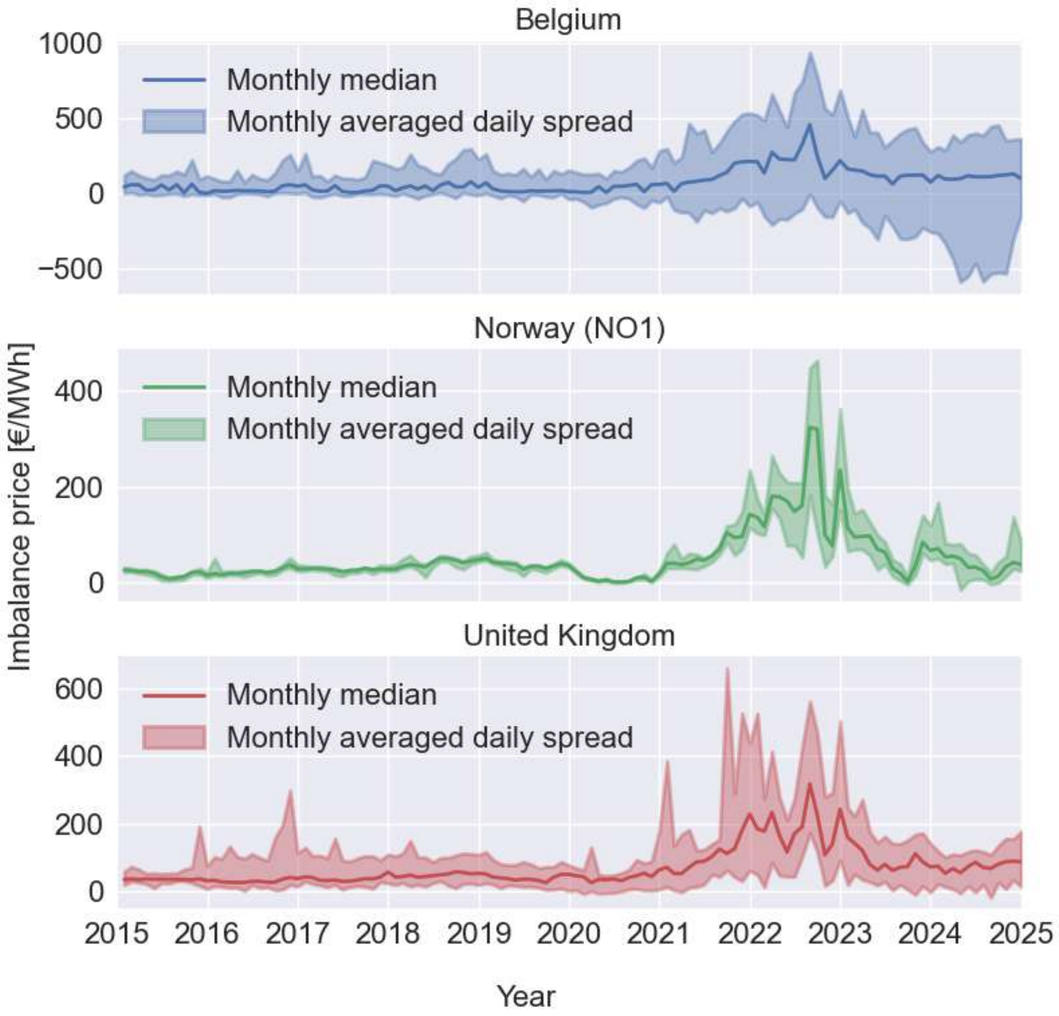
Forecasting the DAM prices in BE and SE(3)

The role of modeling choices (recalibration frequency)



Forecasting the IMB prices in BE

The role of modeling choices (architectures, covariates, …)



Forecasting the IMB prices in BE

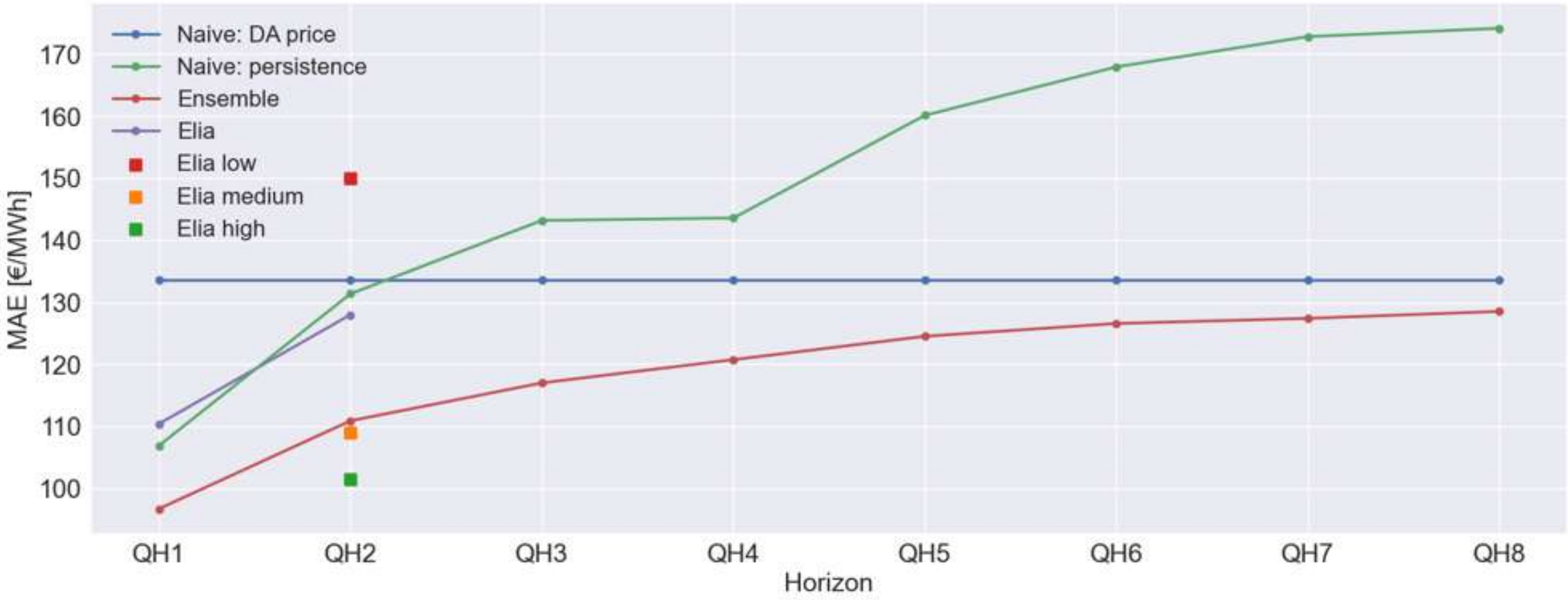
Complexity arising from heterogeneous ‘market’ design

Region	Intraday GCT	Imbalance price system	Imbalance settlement period
Belgium	0 min.	Single as of 01/01/2020	15 min.
France	0 min.	Dual	15 min. as of 01/01/2025
Germany	5 min.	Single	15 min.
Italy	60 min.	Dual	15 min. as of 01/01/2025
Netherlands	0 min.	Dual	15 min.
Nordics	60 min.	Single as of 01/11/2021	15 min. as of 22/05/2023
Portugal	60 min.	Dual	15 min. as of 01/04/2025
Spain	60 min.	Dual	15 min. as of 01/12/2024
United Kingdom	60 min.	Single as of 05/11/2015	30 min.

BRP imbalance	System imbalance Long	System imbalance Short
Long	TSO pays BRP $p_{\downarrow} - \alpha_1$	TSO pays BRP $p_{\uparrow}$
Short	BRP pays TSO $p_{\downarrow}$	BRP pays TSO $p_{\uparrow} + \alpha_2$

Forecasting the IMB prices in BE

The role of modeling choices (architectures, covariates, …)



Concluding remarks

The long road ahead

- Forecasting and scheduling renewables is an increasingly challenging problem, with grave implications for security of supply
  - Upstream weather forecast uncertainty is increasingly driving generation uncertainty in Europe
  - Generation uncertainty is coupled with market uncertainty, and detracts from cost-optimal scheduling of available resources
- Better observability, data and forecast algorithms (incl. in upstream weather forecast models) holds the promise to vastly improve the current status quo